

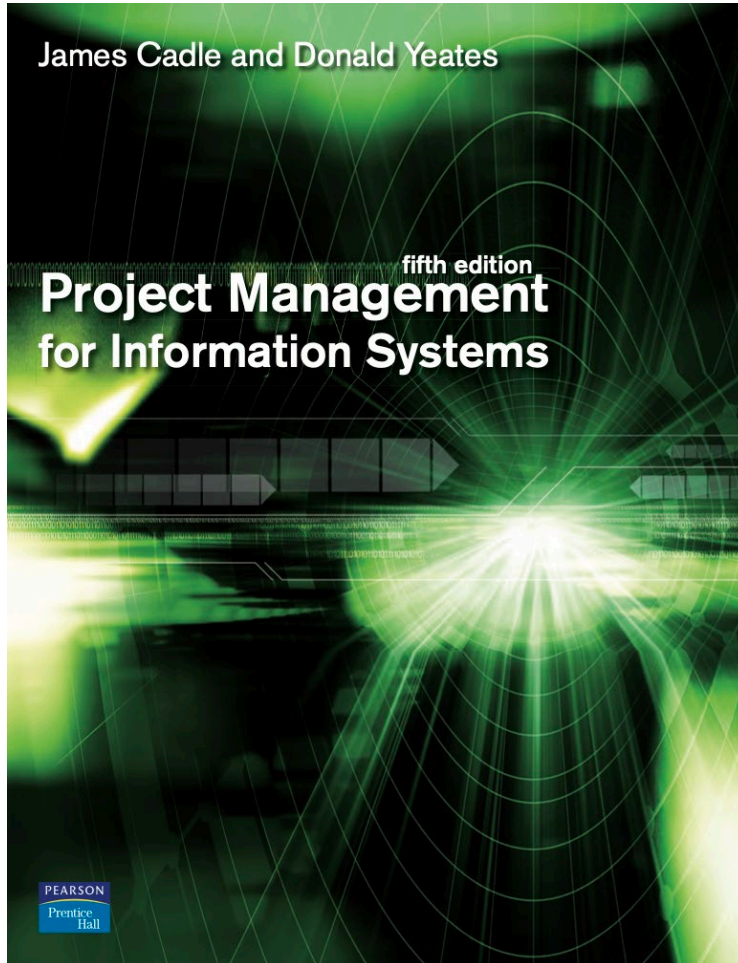
BU5300 – IT Project Management

IT Project Management

Lecture 11: **Summary**

Overview of the Course

Lectures and Book Chapters



Lecture 1 - Chapters 1 and 2

Lecture 2 - Chapters 3 and 4

Lecture 3 - Chapter 6

Lecture 4 - *Agile Development*

Lecture 5 - Chapters 7 and 8

Lecture 6 - Chapter 9 and 10

Lecture 7 - Chapters 11 - 14

Lecture 8 - Chapter 15

Lecture 9 - Chapters 18 and 21

Lecture 10 - Chapter 20

Project management overview, traditional and agile approaches

Chapters 1 and 2

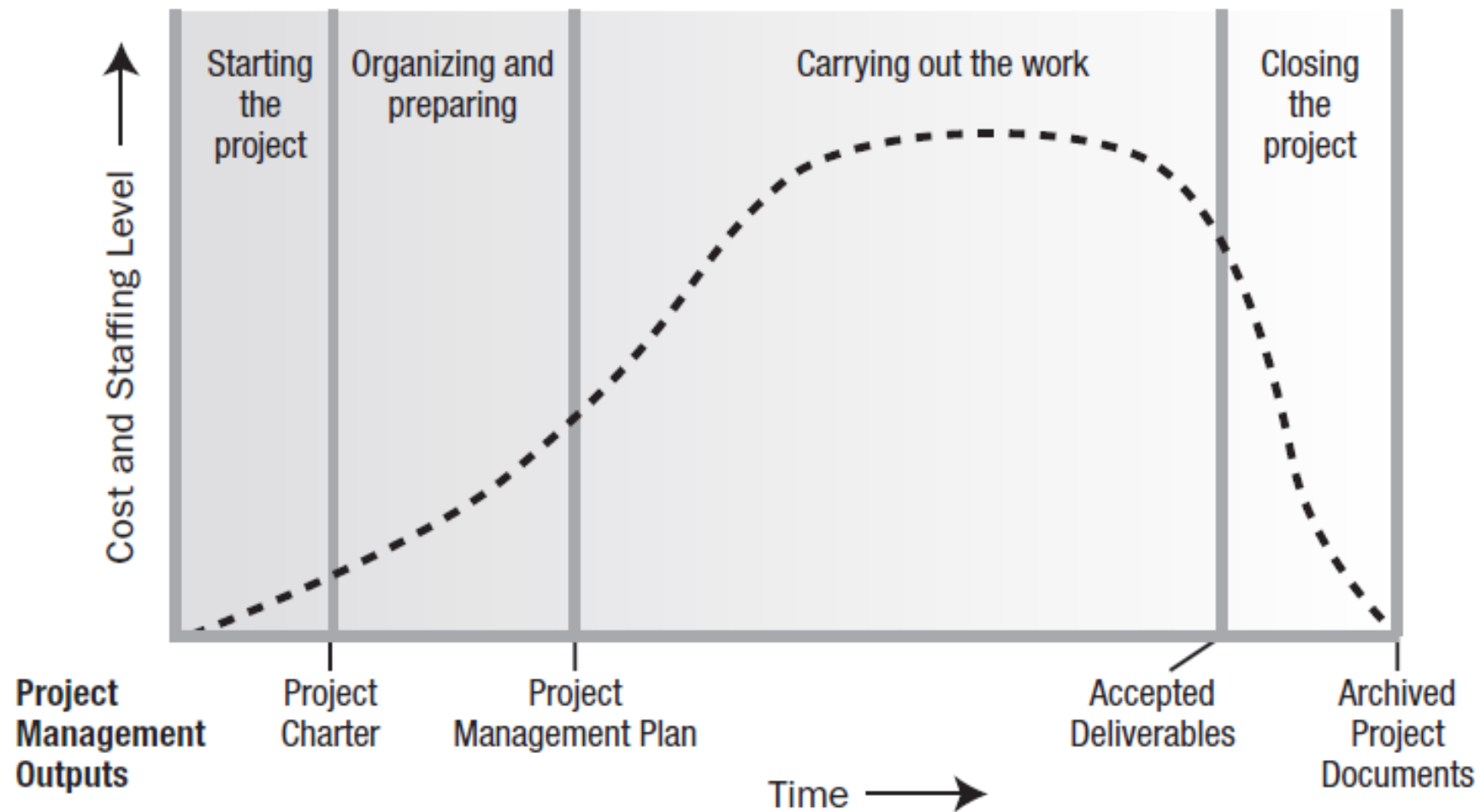
Definitions

by the Project Management Institute (PMI)

- **Project** is a temporary group activity designed to produce a unique product, service, or result
- Project characteristics
 - has a start and an end
 - has an organization and steering committee
 - has a clear mandate and goals for what to produce – often referred to as ‘deliverables’
 - has a defined plan of activities, budget, and schedule
- **Project management** is the use of specific knowledge, skills, tools and techniques to deliver something of value to people.
- The **Project Manager** is responsible for managing the project in terms of plans, budgets, resources, competencies, and communication with relevant actors (e.g., steering committee, functional manager)

Project life cycle

by the Project Management Institute (PMI)

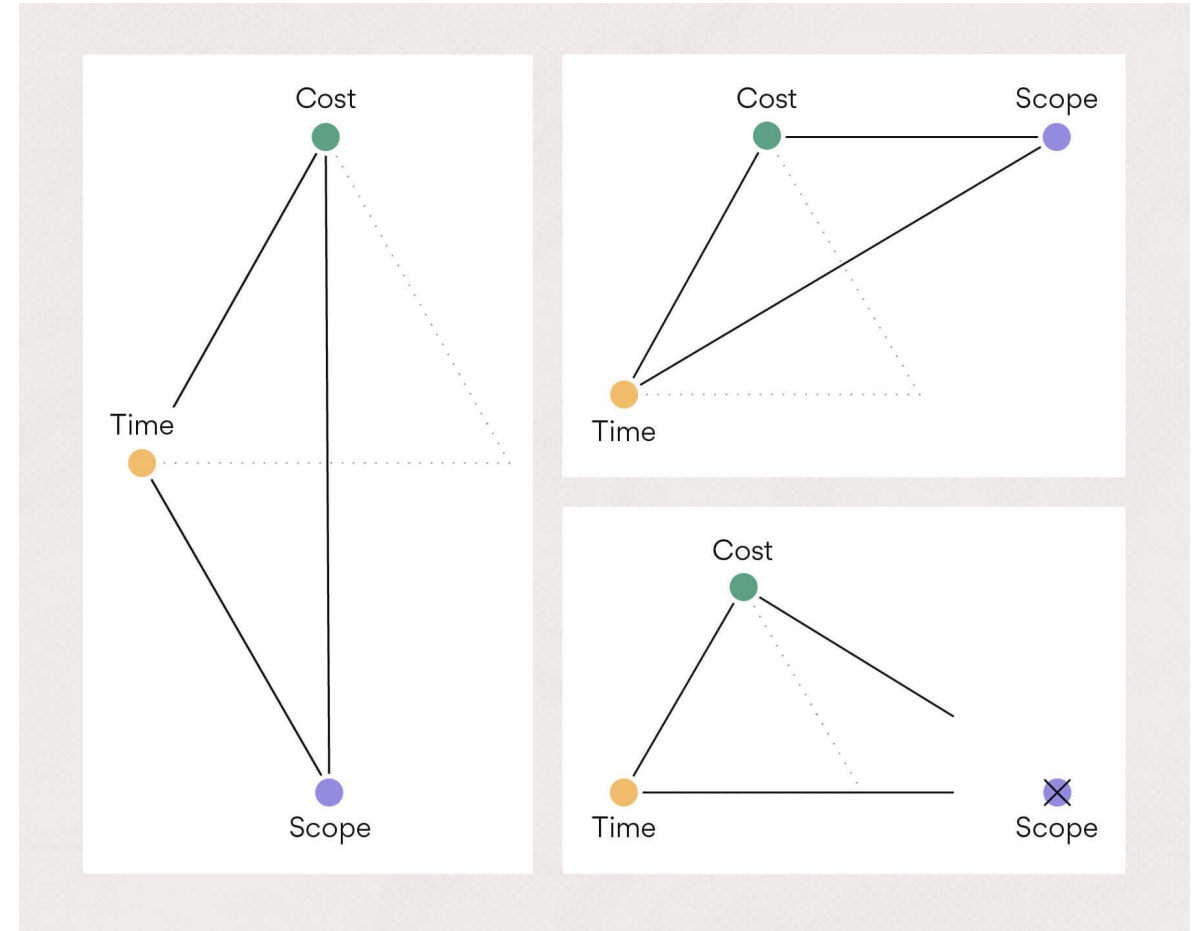


Project life cycle breakdown



Time / cost / scope triangle or “working with triple constraints”

- Project management must balance scope, time, and cost
- Changing one aspect has consequences for the others



Types of information systems projects

1. Software development
2. Package implementation
3. System enhancement
4. Consultancy and business analysis assignments
5. Systems migration
6. Infrastructure implementation
7. Outsourcing (and in-sourcing)
8. Disaster recovery
9. Smaller IS projects

Some IS projects embrace more than one of the nine categories

For example, a project may involve a package implementation, original software development and putting in a new infrastructure

Business Case and Organizational Framework

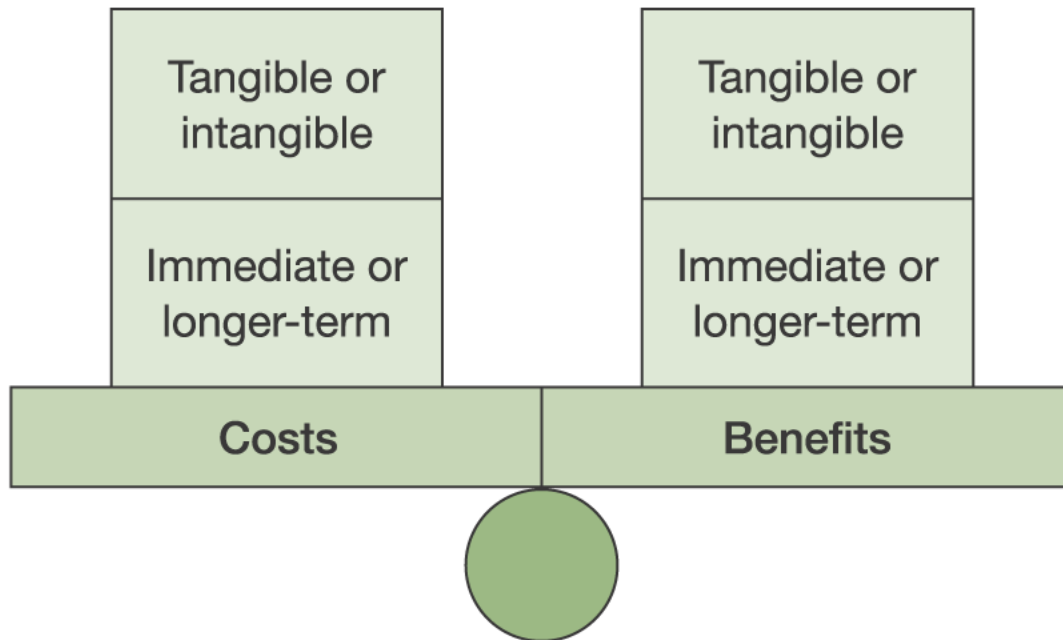
Chapters 3 and 4

Business case

Project Business Case Example			
Project Name	Sales Team IVR Telephone System		
Project Sponsor	Head of Sales	Project Manager	Name of project manager
Date of Project Approval	3rd March	Last Revision Date	3rd March
Contribution to Business Strategy	Our strategy is to project best in industry customer service, and the current situation does not reflect this. The new IVR system will ensure all calls are answered in a timely manner. It will also ensure that calls are delt with efficiently. These two facts align this project to the company strategy.		
Options Considered	Options considered included: 1. Adding additional staff to sales team 2. Having a dedicated team for our best customers 3. An IVR system (selected)		
Benefits	1. Increased sales - currently estimated we lose 4% of all sales calls due to current issues. 2. Happier customers - we estimate new customer satisfaction will increase by 10%. 3. Improved LTV - lifetime value of customers will increase by 5% due to the two points above		
Timescales	Initial analysis shows that the system will take approximately 3-4 months to implement.		
Costs	IVR software = \$35,000 Project Management = \$30,000 Software team of 3 for 3 months = \$90,000 Total estimated cost = \$155,000		
Expected Return on Investment	Year 1 = \$0 Year 2 = \$120,000 Year 3 = \$180,000 as LTV begins to be felt.		
Risks	Right now the project looks pretty straightforward but there are still some unknowns surrounding implementation. There is also the risk that the project doesn't meet the sales team or customers needs. For this reason it is recommended to involve the sales team closely.		

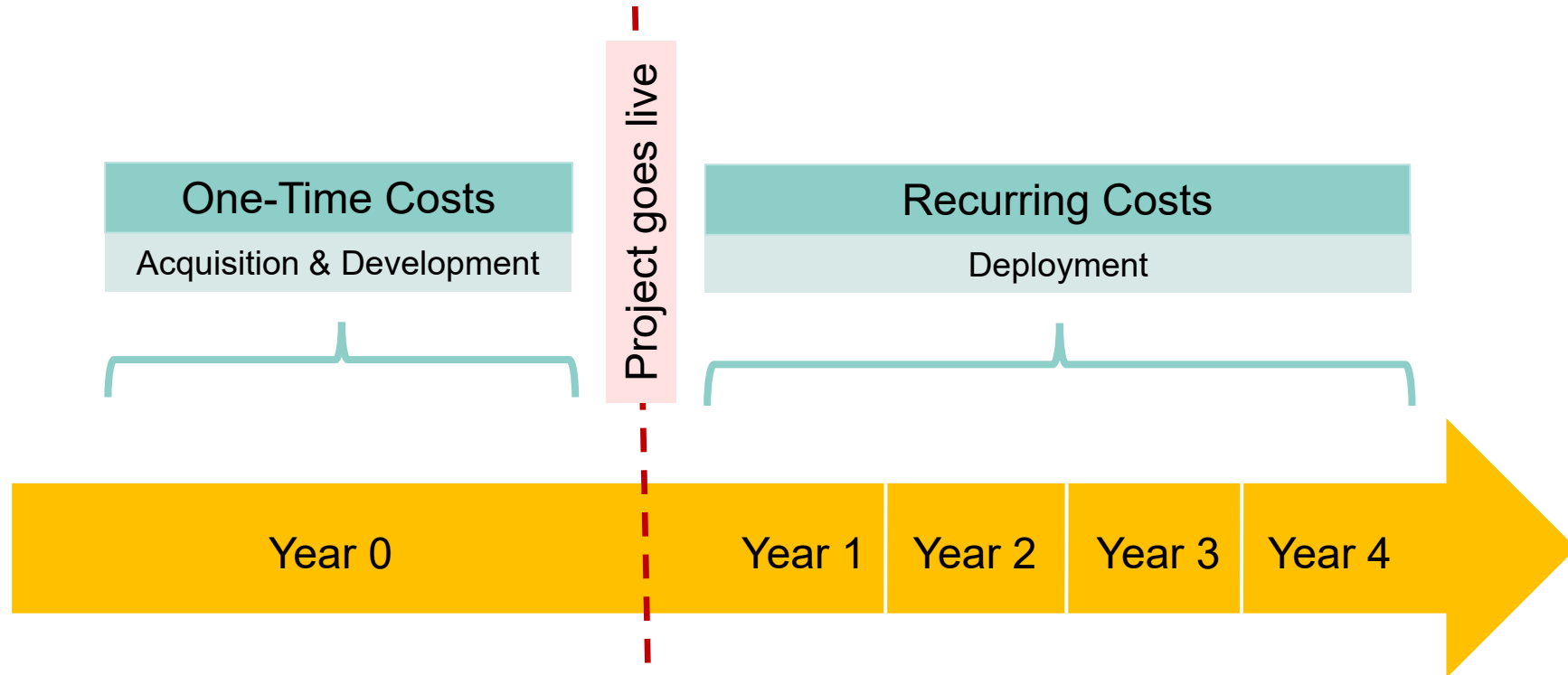
- Introduction and background
- Management summary
- Description of problem or opportunity
- Options available and considered
- Cost/benefit analysis
- Impacts and risks
- Conclusions and recommendations

Costs and benefits



- **Tangible** (which means they can be plausibly quantified in some way) or
- **Intangible** (which means that they cannot be so quantified)
- **Immediate / long term** realization

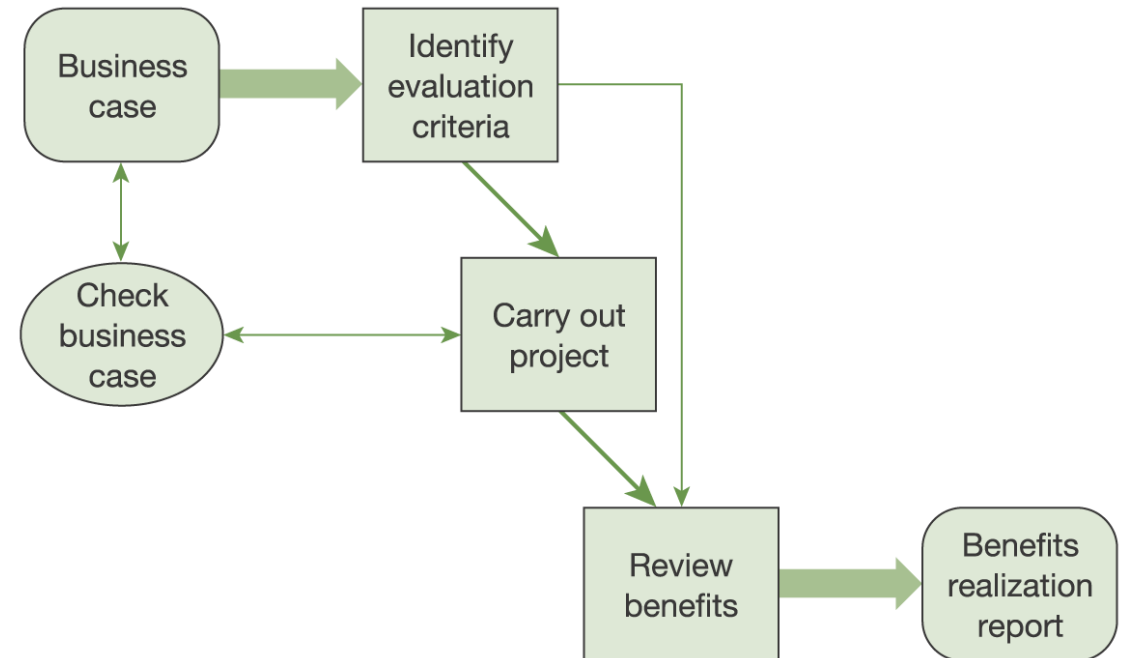
IT Project Costs



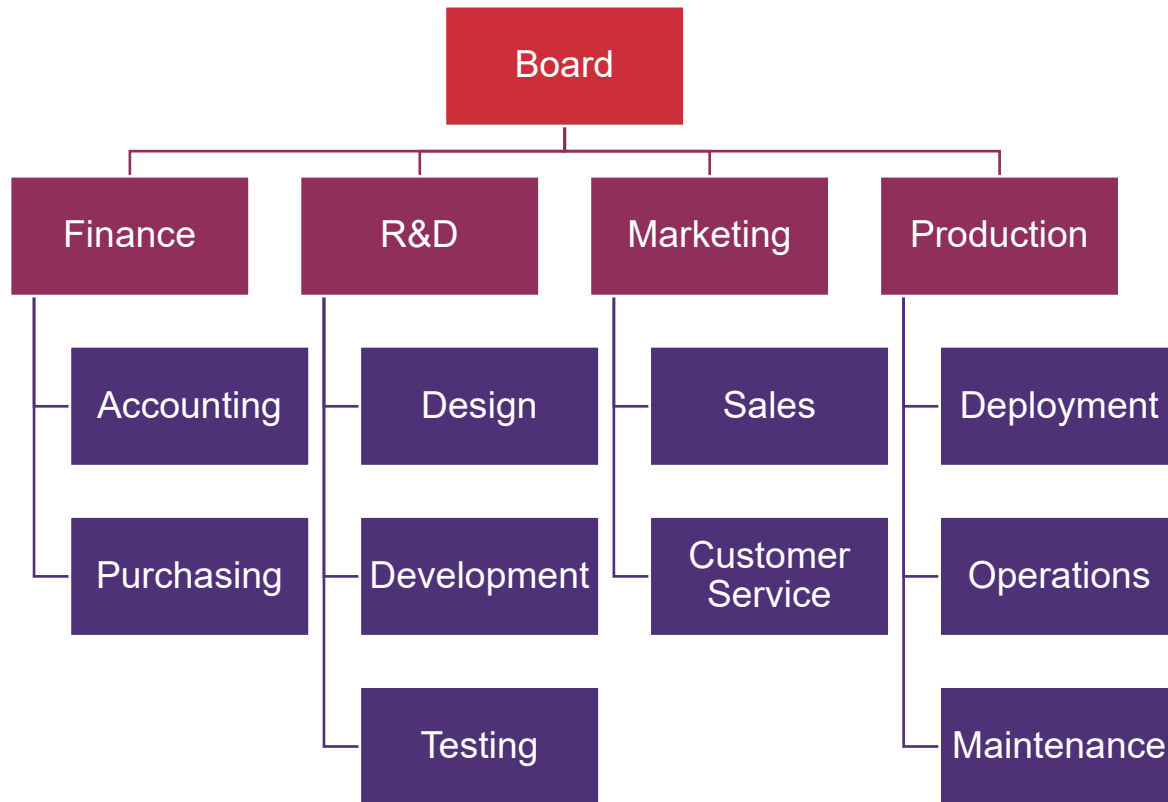
Return of Investment

- Payback calculation – simple and straightforward
- Discounted cash flow:
 - Takes into account ‘time value of money’
 - Produces a net present value for project
 - Sensitivity analysis can be applied
- Internal rate of return – produces single headline number for comparisons

Benefits realization

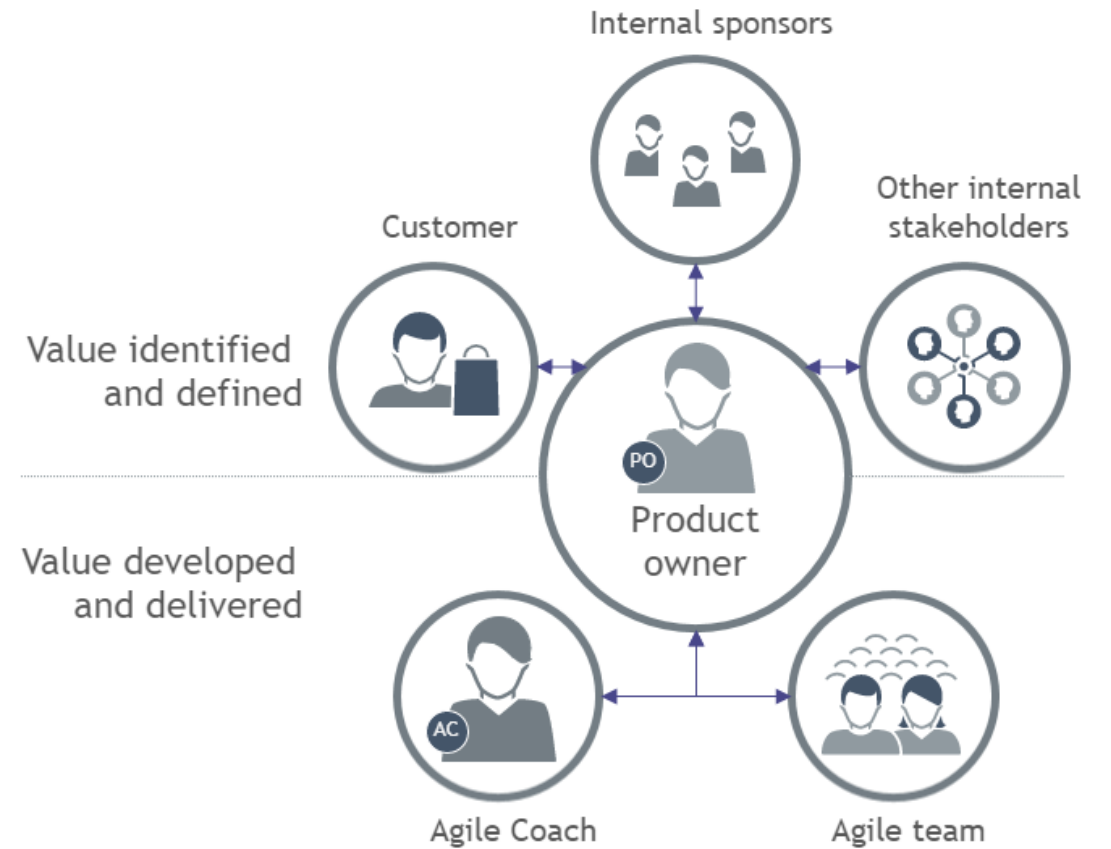
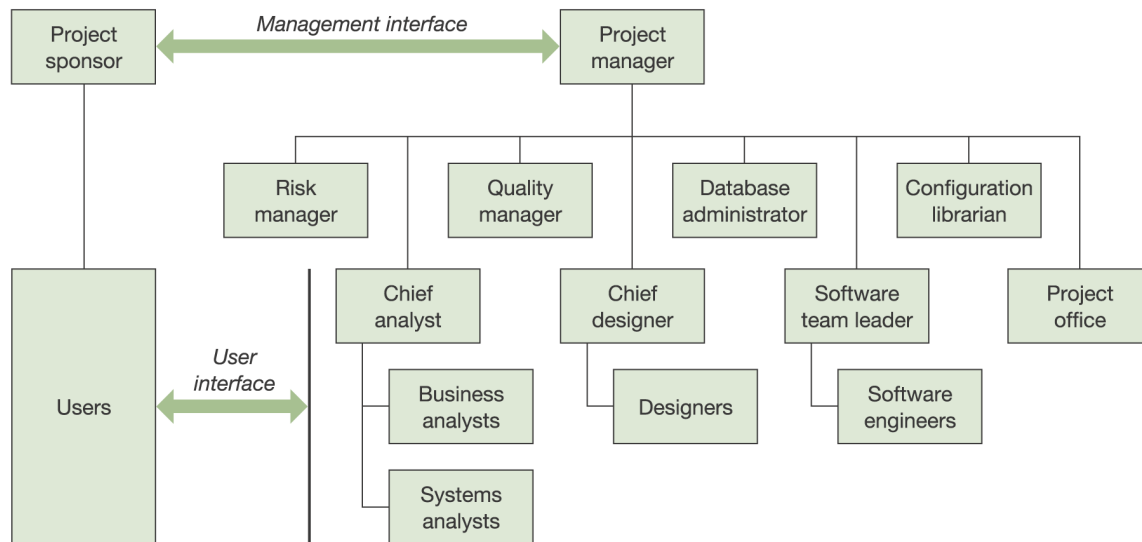


Functional organization



- The organization is divided into departments based on specialization
- Advantages:
 - Encourage the development of a high degree of specialist skills
 - People in the functions report to managers who understand their fields
- Disadvantages
 - can develop a '**silo**' mentality – being more concerned with departmental objectives than with the needs of the organization as a whole

Project organization and roles



Sponsor vs Project Manager

- Define business aims
 - Justify project to board
 - Define project objectives and priorities
 - Specify minimum requirements for success
 - Obtain capital expenditure approval
 - Monitor project progress from business standpoint
 - Keep higher management informed of progress
 - Provide high-level support as project champion
- Achieve project objectives within constraints
 - Make timely decisions for success
 - Plan, monitor, and control project
 - Build and motivate project team
 - Keep sponsor and management informed
 - Recommend project termination if necessary
 - Serve as principal point of contact
 - Manage subcontractors

Development Lifecycles

Chapter 6

Traditional vs Agile approaches

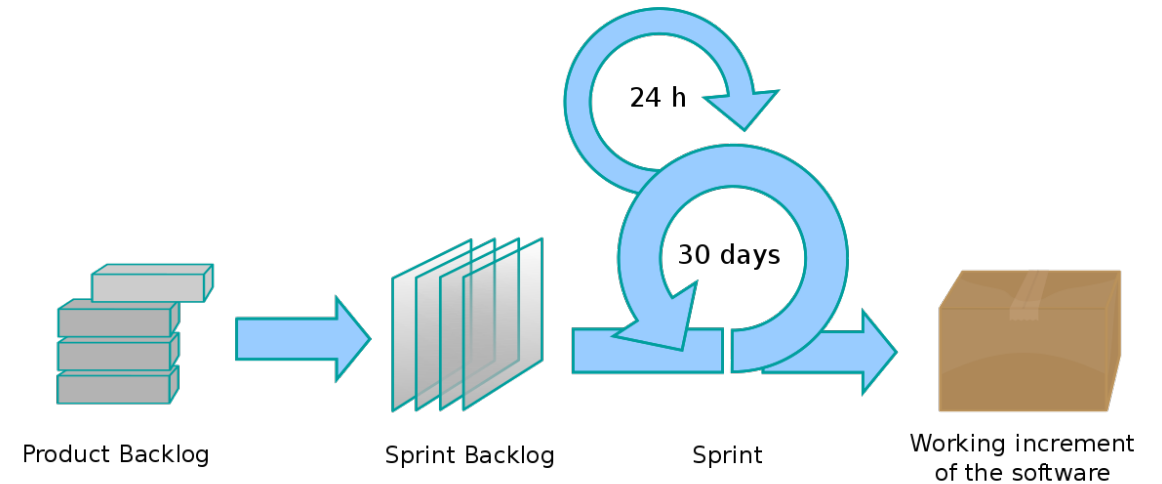
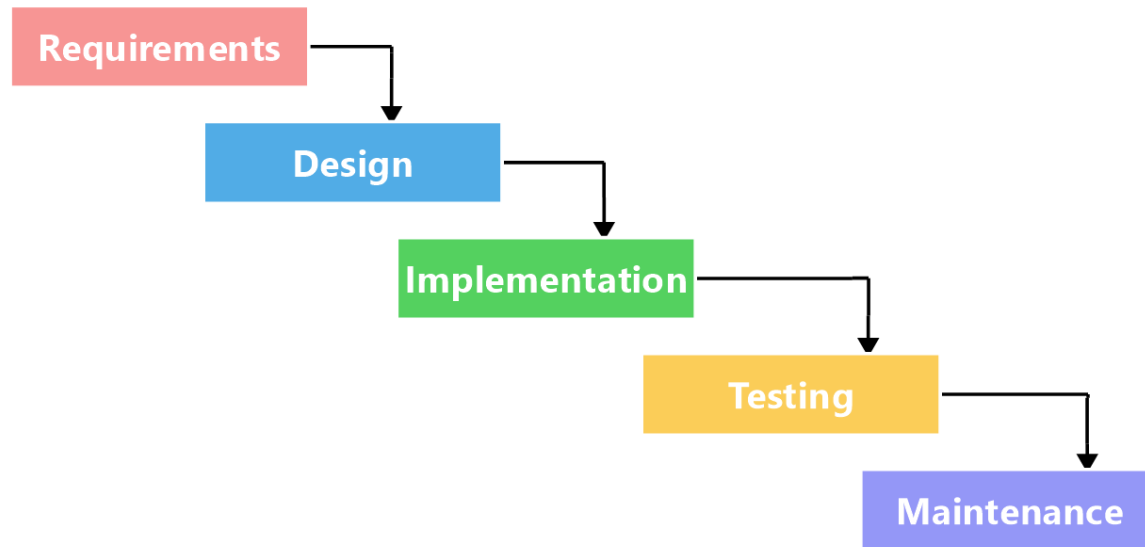
Traditional

- Focus on management control and planning
- Detailed plans carried out in a waterfall-like fashion
- Hierarchical organizations often with a top- down approach to project governance
- IT-projects: requirements must be defined up-front and not change too much

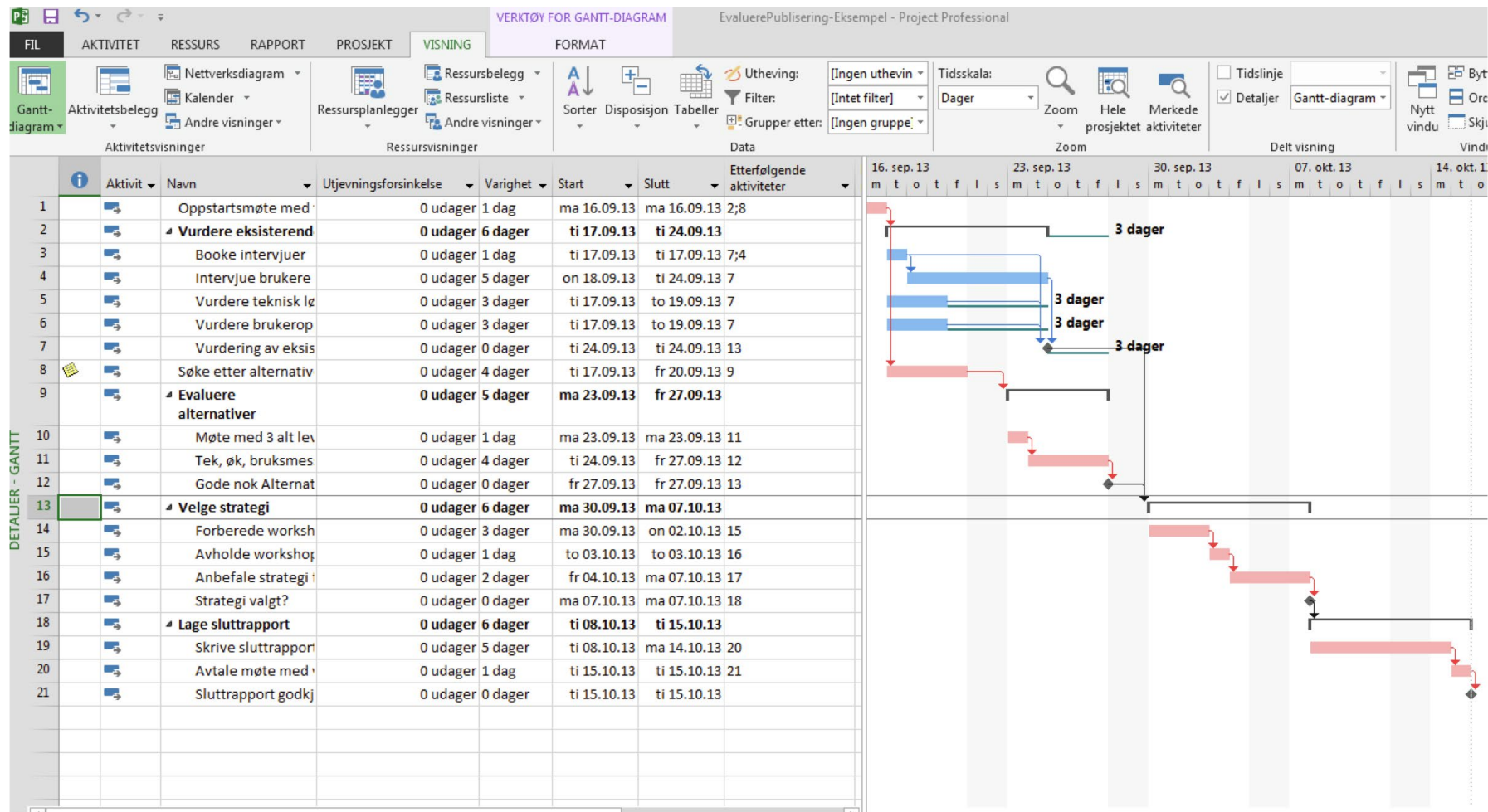
Agile

- Co-located and small teams
- Coordination through physical artifacts as well as software tools
- Close interaction with customer
- Self-managed teams

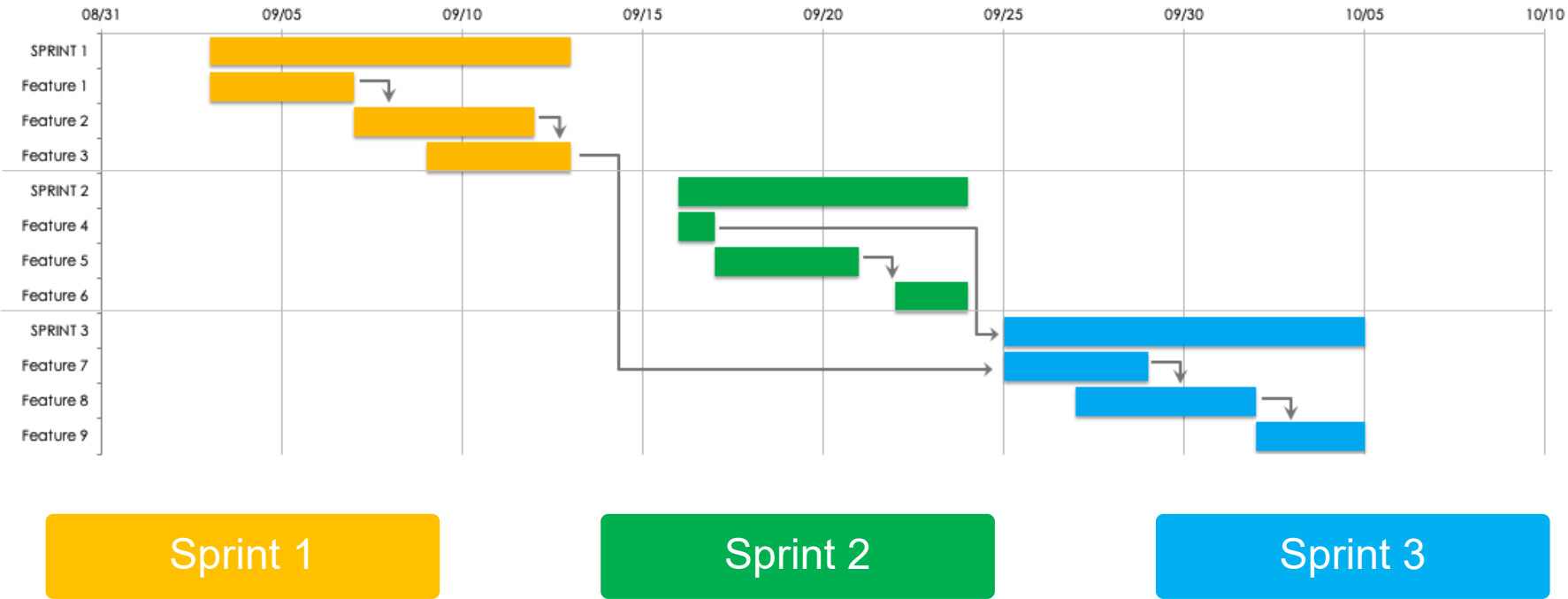
Traditional vs Agile approaches



GANTT chart

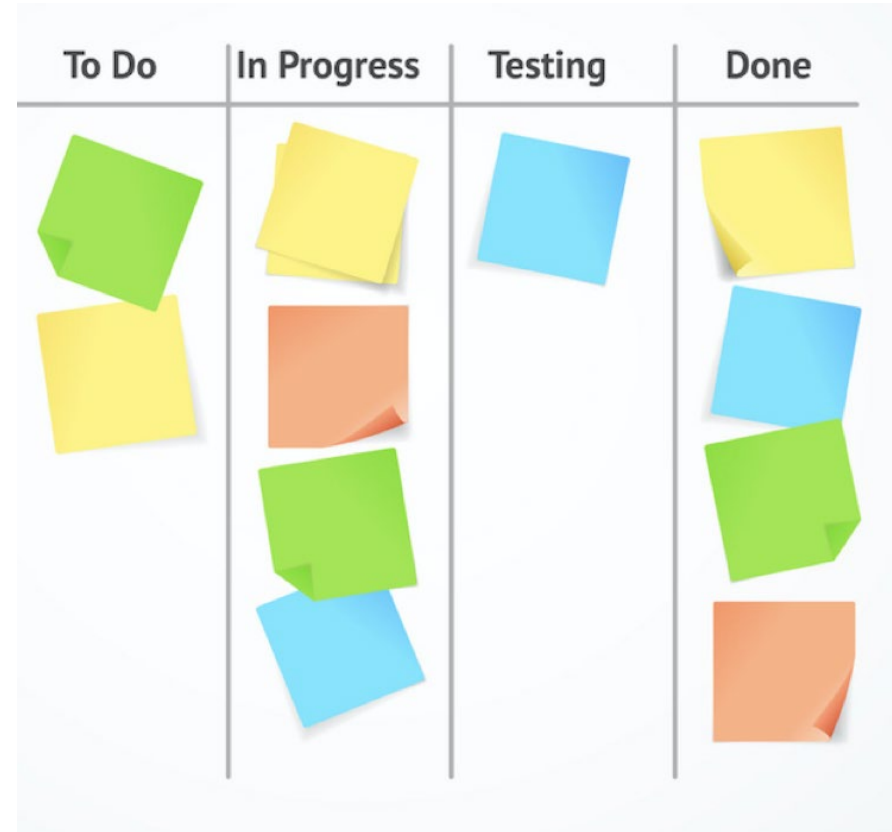


GANTT chart

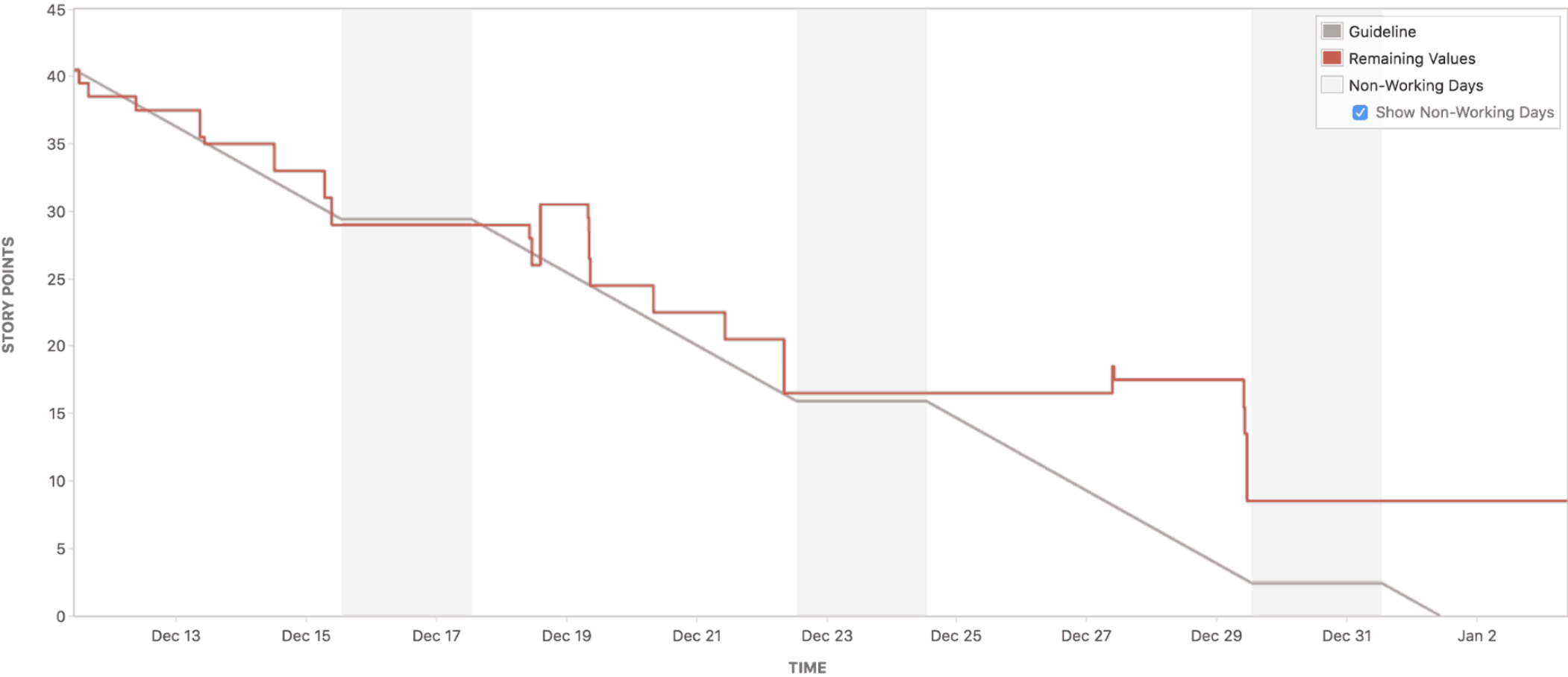


Kanban Board

- It helps visualize work, progress, and assess efficiency
- Every work item is represented as a separate card on the board
- As work is progressing, team members move the cards to the right columns



Burn-down charts



Project Initiation and Project Planning

Chapters 7 and 8

- The process under which the project is created and defined **to the extent necessary to begin planning**
 - It puts the project on a solid foundation
 - Ensures that the organization is clear about the purposes and priorities
 - A place from which the project manager and project board can assess progress
 - Its importance is easily underestimated
- PID (also Project Charter) is the document which outlines the organization's vision for the project and the business case for it

Components of a PID

- ✓ Project definition and scope
- ✓ Project approach and milestones
- ✓ Business case and success criteria
- ✓ Funding amount and funding status
- ✓ Project management team
- ☐ Stakeholders
- ☐ Project plan
- ☐ Project controls
- ☐ Quality management
- ☐ Change controls
- ☐ Risk management
- ☐ Communication management

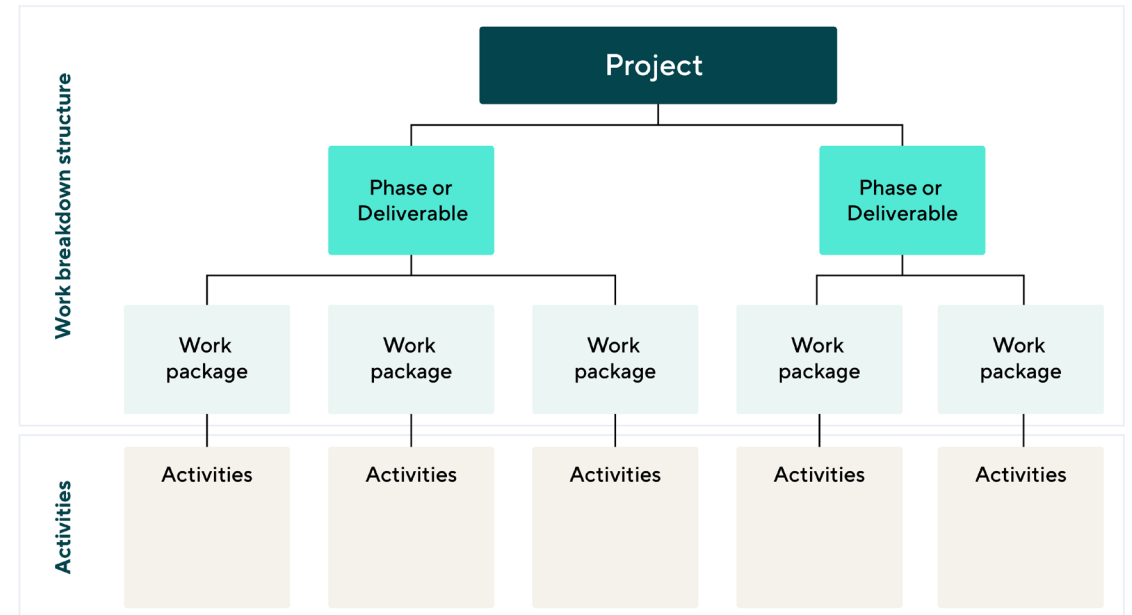
Breaking down the work

Work packages are the smallest unit of work that can be assigned to an individual or a team

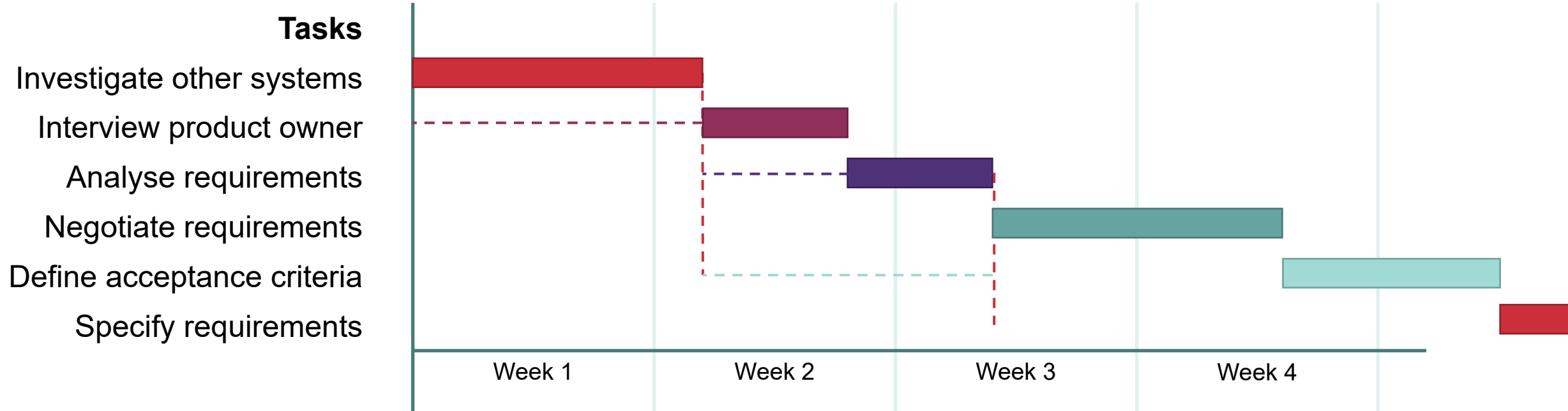
- Description of the work to be done (WBS)
- Deliverables or outputs to be produced (PBS)

Also includes

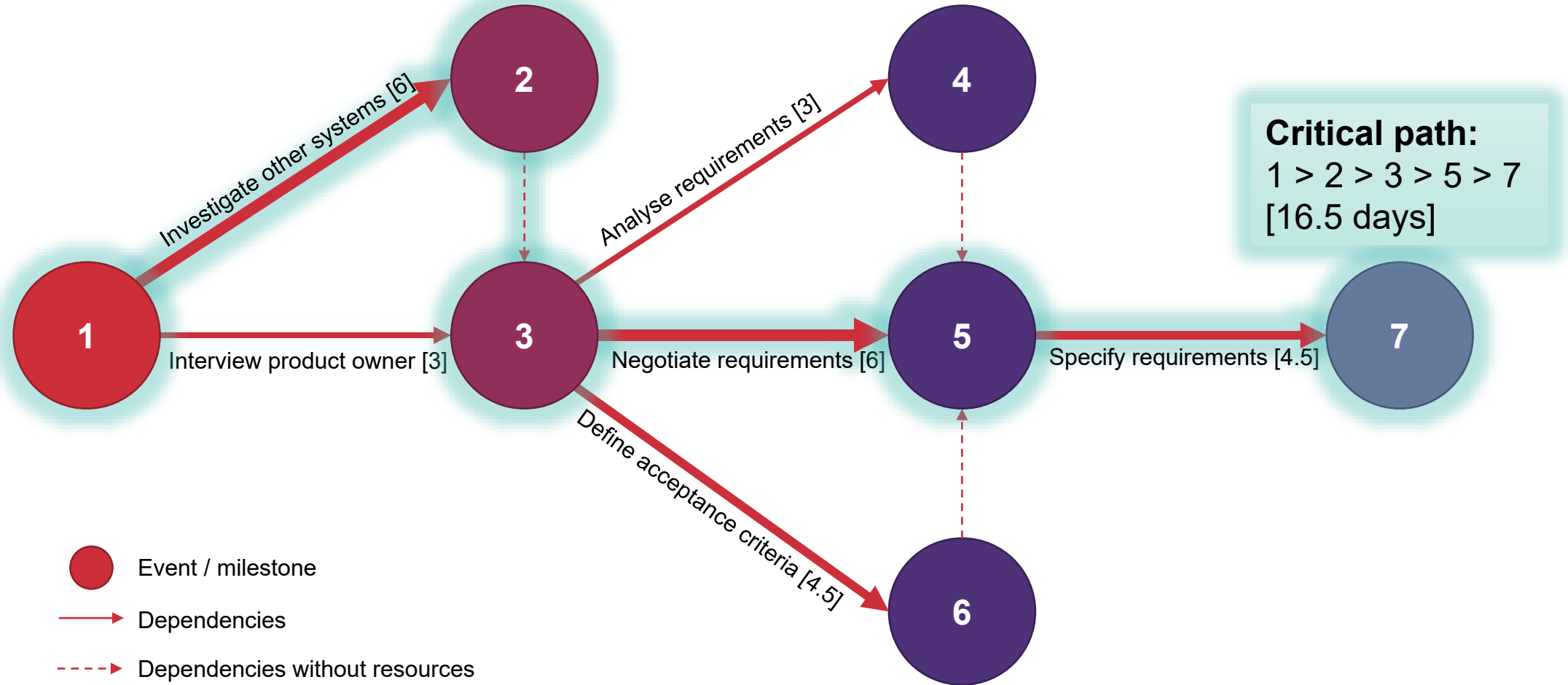
- Timeline/schedule for completing the work
- Resources needed such as people, materials, or equipment
- Budget or cost estimate
- Quality standards and criteria for acceptance
- Risks and strategies for managing them



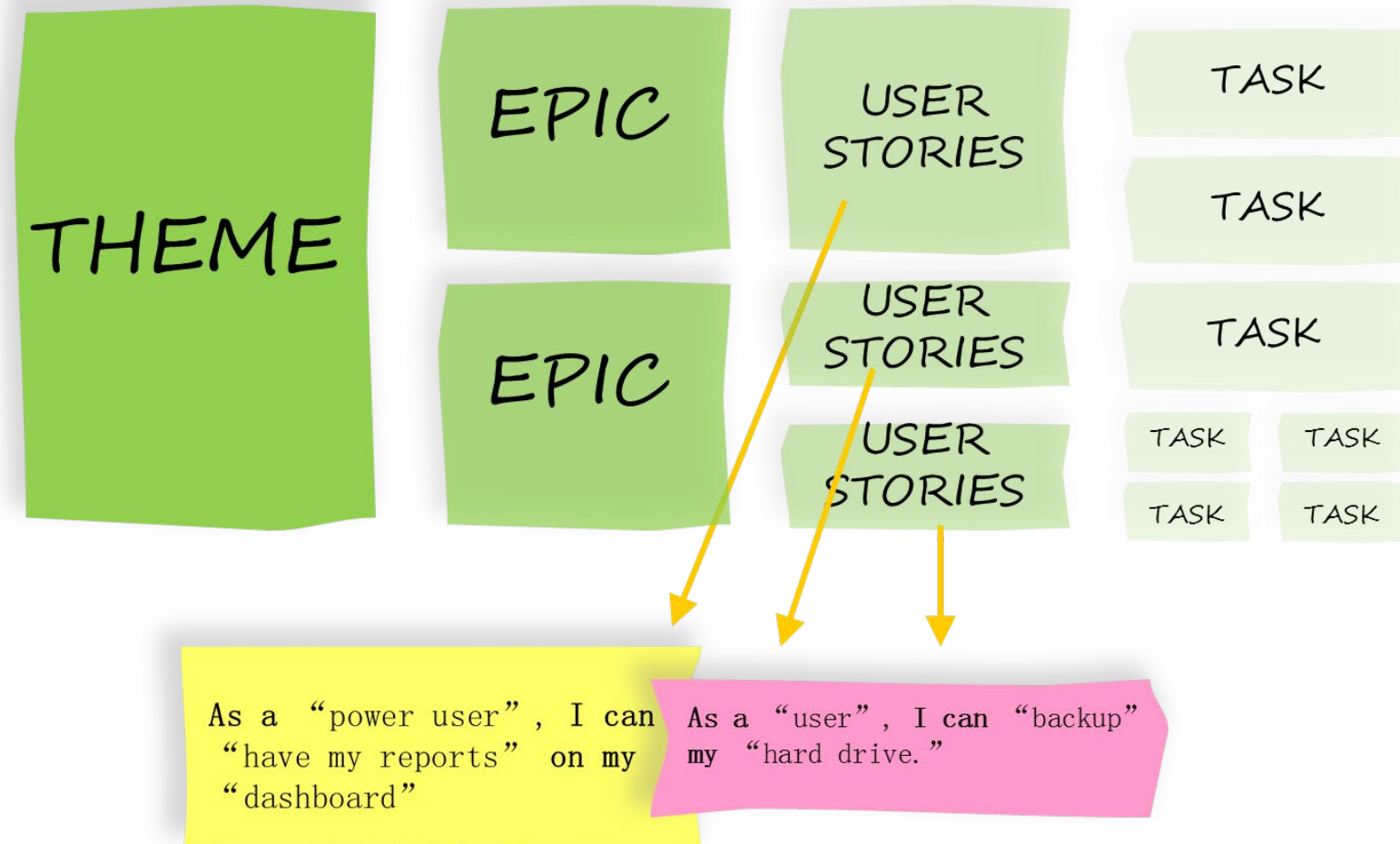
Sequence of activities



Network diagram (PERT chart)



Breaking down the work with Agile

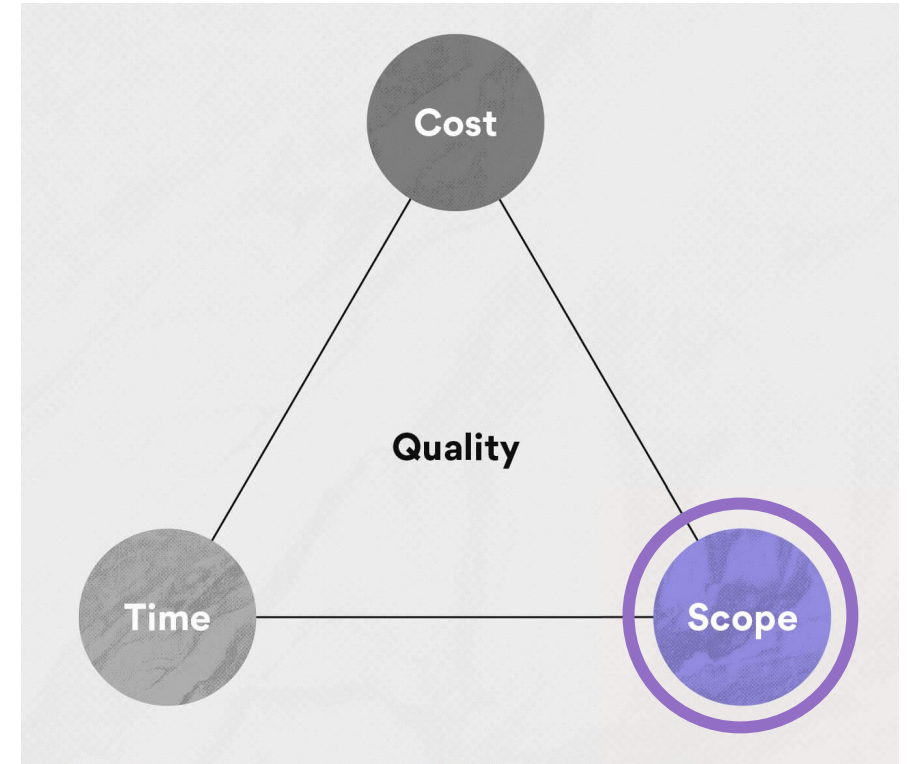


Requirements and Estimation

Chapters 9 and 10

Software Requirement

- A description of something the software must do or property it must have
- The set of system requirements denote the problem to be solved and any constraints on the solution



Functional vs. Non-functional Requirements

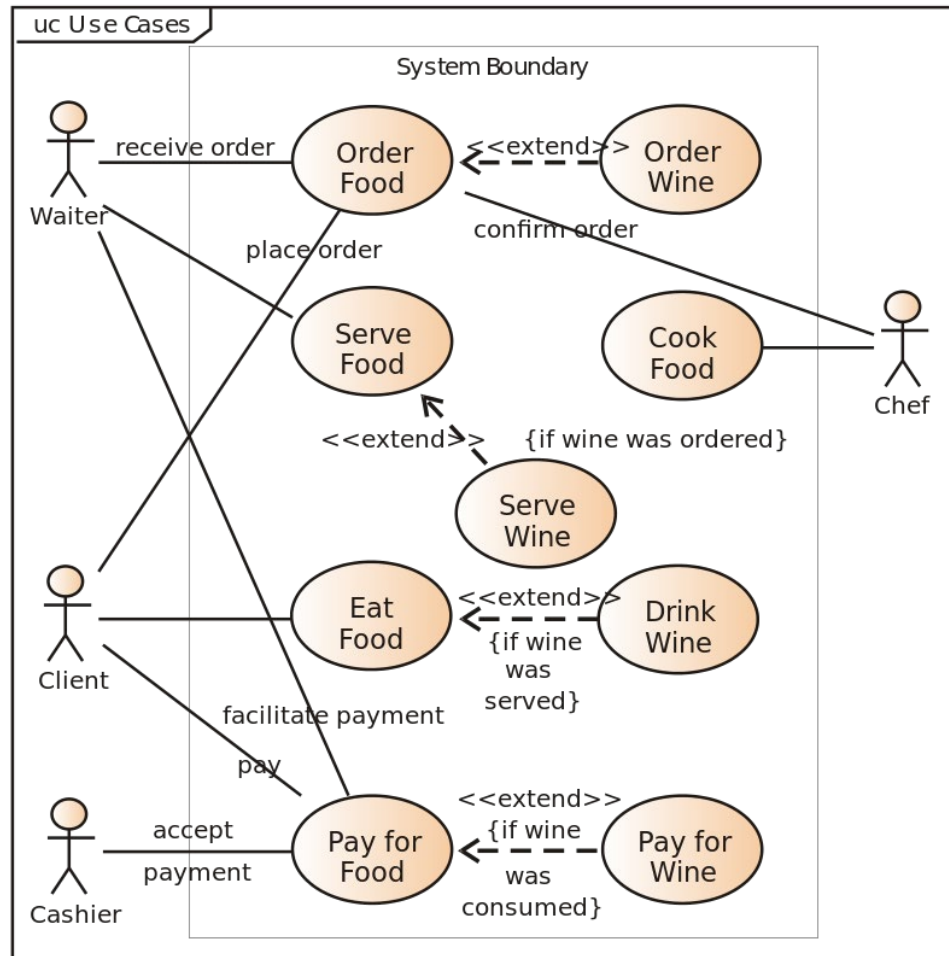
Functional Requirements:

- What the system/application should do
- Describes system behaviors and interactions
- Specific and measurable
- Defines features and functionality
- Examples: User authentication, Payment processing, Reporting

Non-functional Requirements:

- How the system/application should perform
- Describes system qualities and characteristics
- General and subjective
- Defines performance, reliability, and usability
- Examples: Response time, Scalability, Security, User Experience

Use cases vs User stories



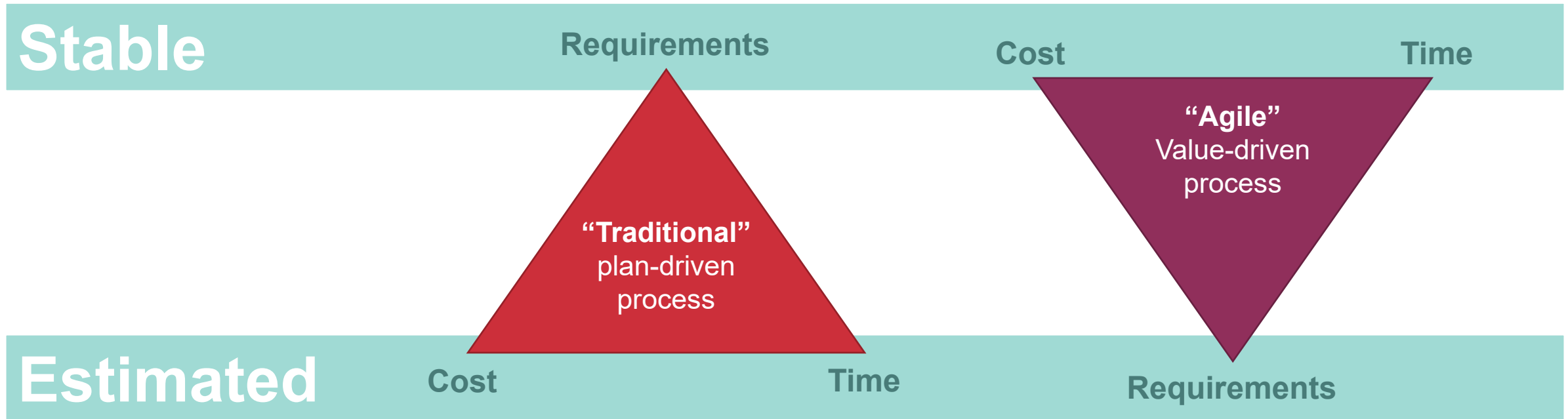
**As a <role>
I want <goal>
so that <benefit>**

Acceptance criteria:
(Conditions of Satisfaction)

...

...

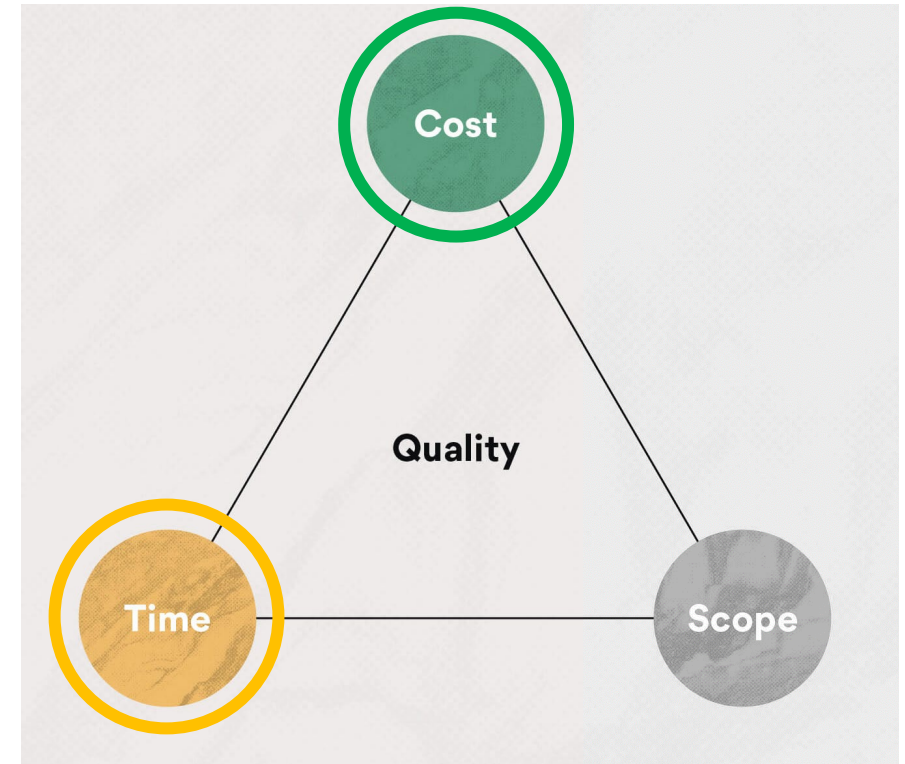
Stable vs flexible requirements



Estimation

An assessment of the size of tasks with **costs, time spent** (hours and calendar time), and the number of people needed to realize the products in the project

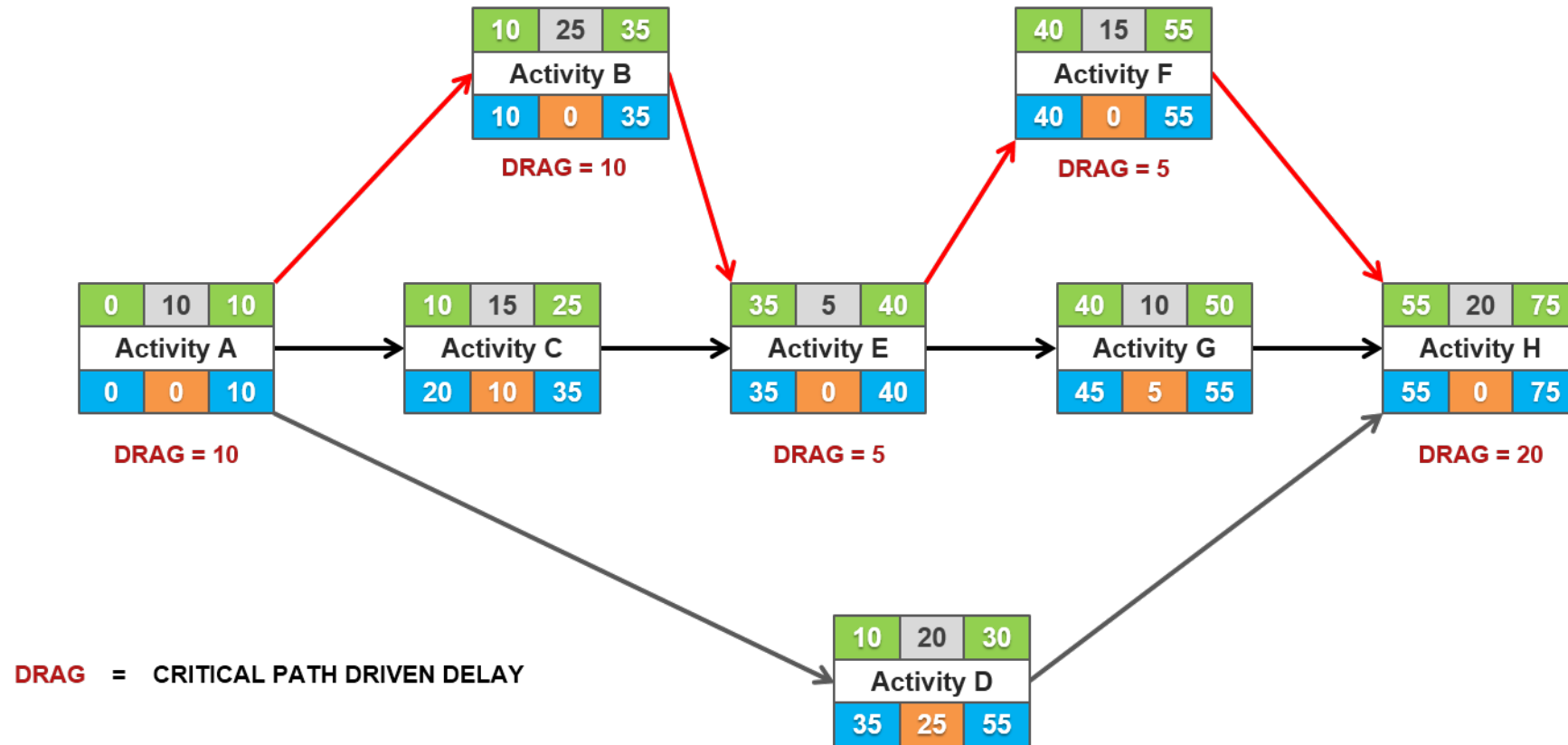
The assessment is based on experience, various empirical and analyst techniques



Many ways

- Estimation by Analogy (EBA)
- Estimate based on scope of analysis phase
- Estimate based on the scope of coding
- Live estimation based on WBS/PBS
- Delphi method
- CoCoMo
- Function Points
- Critical path analysis or PERT estimating
- Jorgensen's simple approach
- Mean magnitude of relative error (MMRE)
- Expert estimation

PERT Estimating



Story Points Estimation

How much we know	Dependencies	How much work effort	Story Points
Everything	None	< 2 hours	1
Almost everything	Almost none	Half a day	2
Something	Some	Up to 2 days	3
Almost nothing	Few	Few days	5
Nothing	More than a few	Around a week	8 – must be split into smaller itens
Nothing	Unknown	More than a week	13 – must be split into smaller itens

Pick a row that represents your task best; if it fits more than one, pick the higher one.

Monitoring, controlling, and reporting progress

Chapters 11 - 13

Monitoring Progress - Overview

- Measuring project progression and performance
- Three perspectives: **time**, **cost** and **quality**
- The first step is to collect data at an individual level

Name: <i>DAVE SIMS</i>		Project: <i>PERSONNEL</i>							Week ending: <i>19 May</i>	
Code		Mon	Tue	Wed	Thu	Fri	Sat	Sun	Total	To go
<i>A/01</i>	<i>CODE PROGRAM CV004</i>	<i>6.5</i>	<i>5.0</i>	<i>7.5</i>					<i>19.0</i>	<i>NIL</i>
<i>A/02</i>	<i>TEST PROGRAM CV004</i>				<i>7.5</i>	<i>3.0</i>			<i>10.5</i>	<i>NIL</i>
<i>A/07</i>	<i>CODE PROGRAM EN025</i>					<i>4.0</i>			<i>4.0</i>	<i>15.0</i>
<i>M/03</i>	<i>TEAM MEETING</i>	<i>1.0</i>							<i>1.0</i>	
<i>M/02</i>	<i>COMPLETE TIMESHEET</i>					<i>0.5</i>			<i>0.5</i>	

Monitoring Time (Effort on Project-Level)

- Summarize it at a project level to see the overall situation
- Project planning packages usually have facilities to do this
- For each activity, we need to know:

Not Started (planned)	In Progress (current or estimated)	Completed
Planned start and end date	Start date <i>Did we start in time?</i>	End date <i>Did we delivered in time?</i>
Effort and cost estimates	Effort and cost (done + still to go) <i>Are we on the right track?</i>	Total effort and cost done <i>Are our estimates precise?</i>

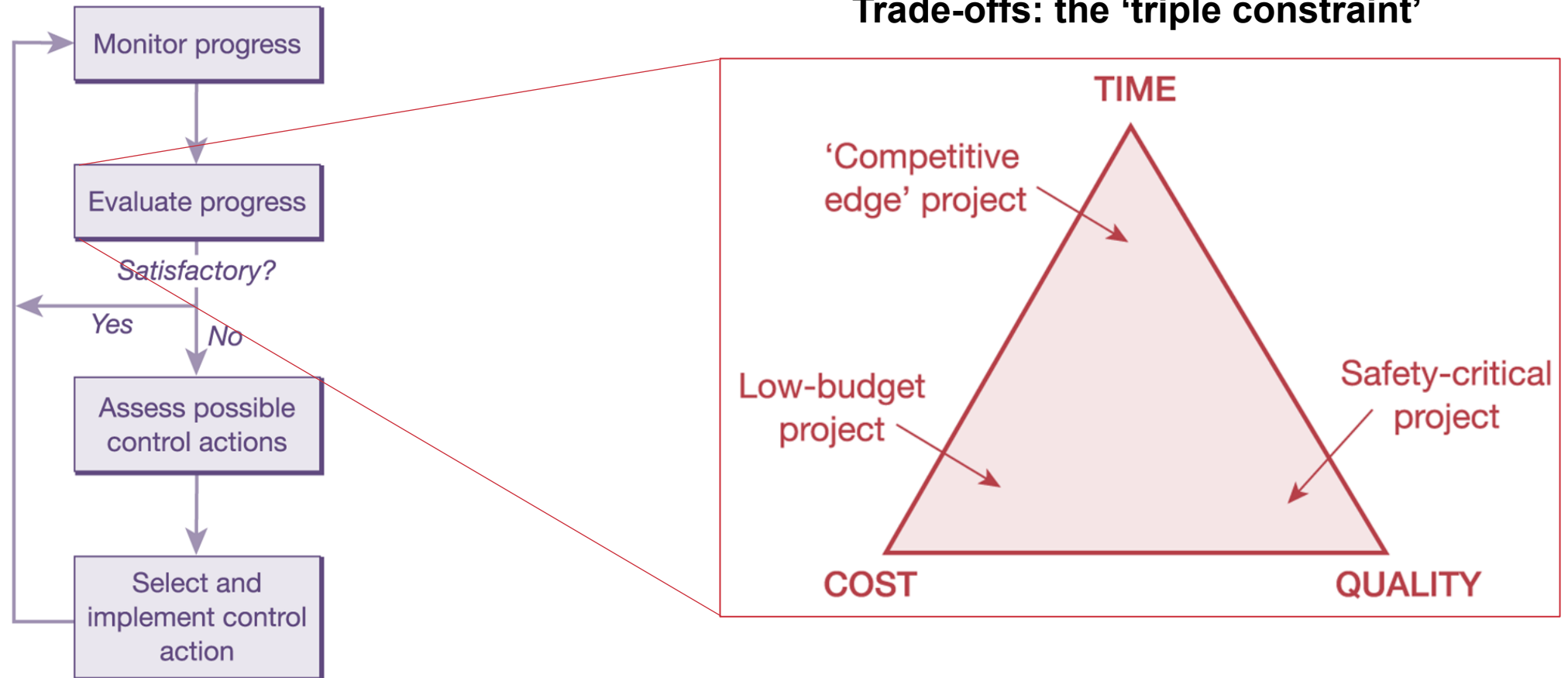
Monitoring Quality

- Criteria that **a particular task or user story** must be met to be considered complete
- Typically includes a set of objective standards and guidelines
 - The team should create a checklist that must be passed before delivery
- Helps ensure that work is completed to a consistent standard of quality and completeness

DoD Checklist example

- ☐ Product owners accepts the User Story
- ☐ Passed unit testing
- ☐ Code review
- ☐ Meet acceptance criteria
- ☐ Passed functional testing
- ☐ Meet non-functional requirements

Control cycle



Change control and configuration management

Change control

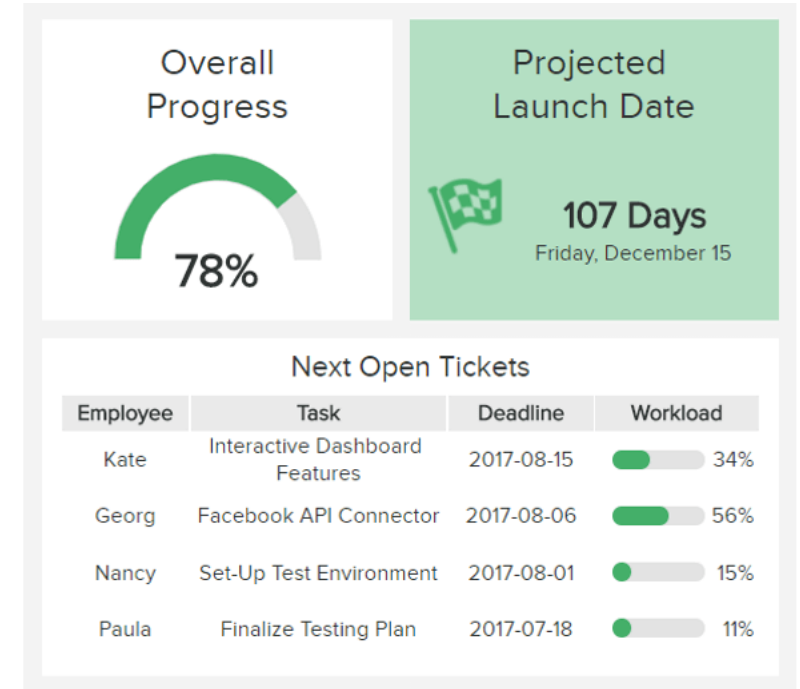
- Set of procedures to ensure that changes made only after a consideration of the full impacts
- Organizational tools to help individuals make successful transition
- Project perspective

Configuration management

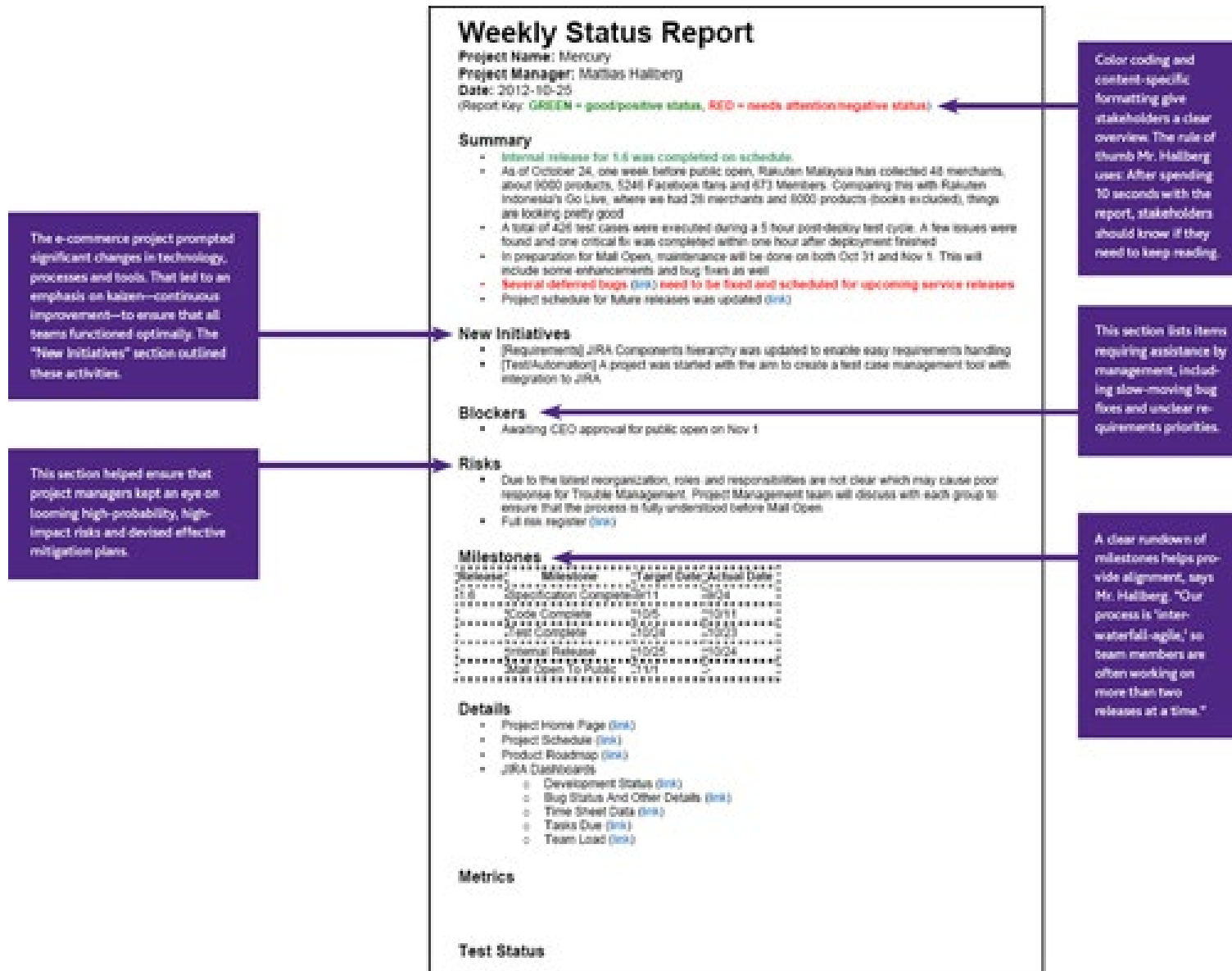
- Version control to ensure that all changes are properly recorded and managed
 - and so that knock-on effects on other projects can be identified
- Identify and monitor changes on the functional characteristics
- Product perspective

Report contents (typical)

- Period covered
- Narrative summary of progress
- Milestones achieved/deliverables completed
- Problems encountered (and solutions)
- Projected completion date
- Costs to date and predicted
- Changes identified and implemented

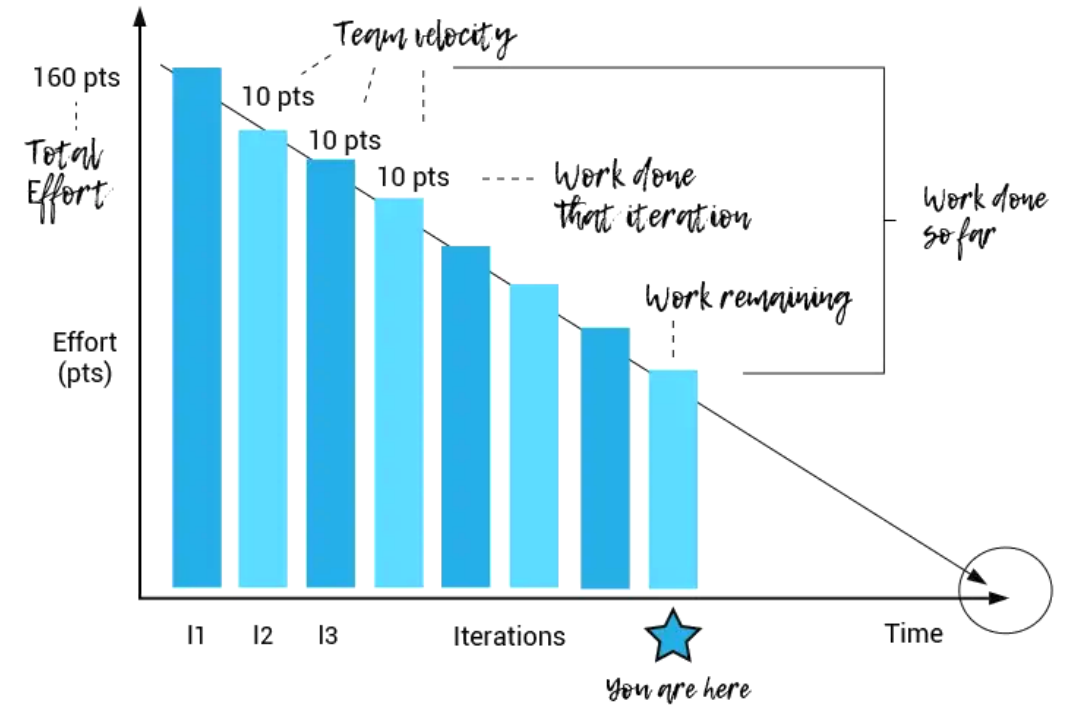
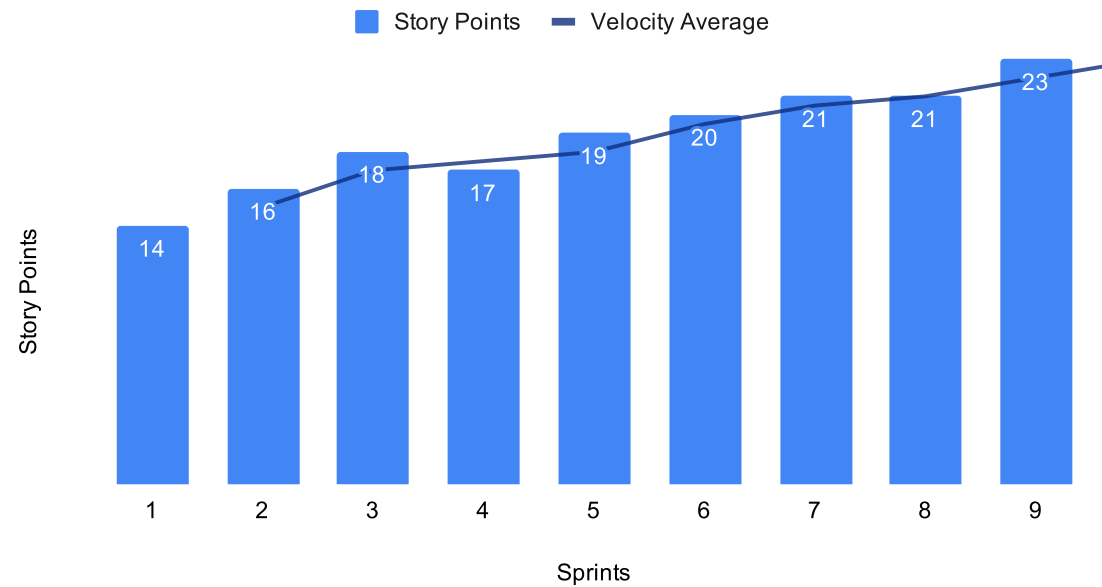


Status Report



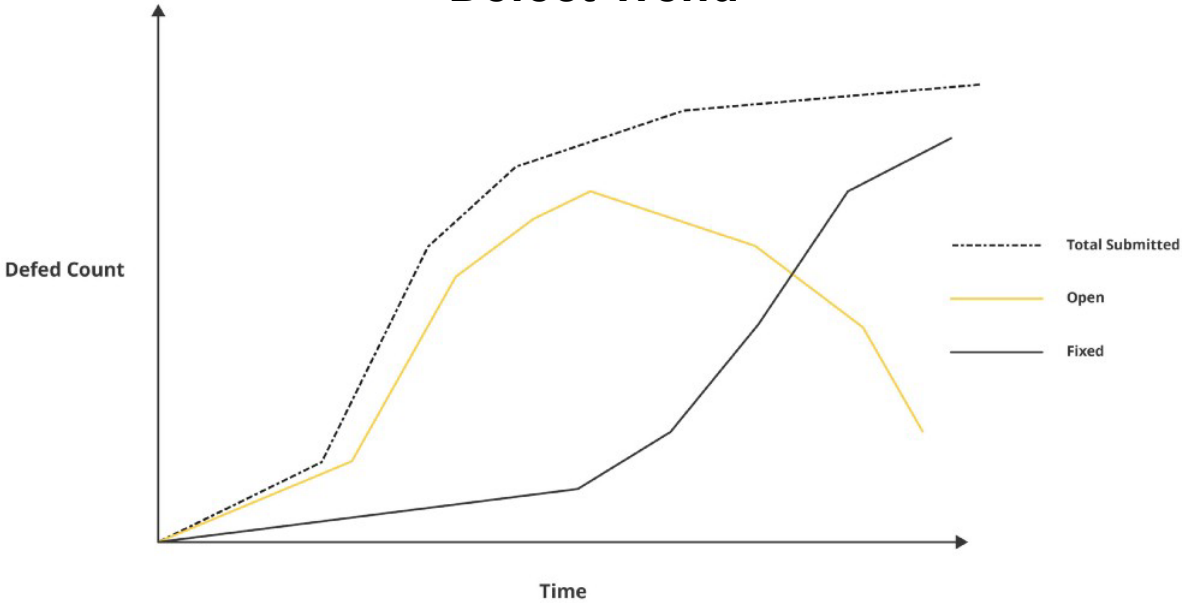
Progress in Agile

Velocity

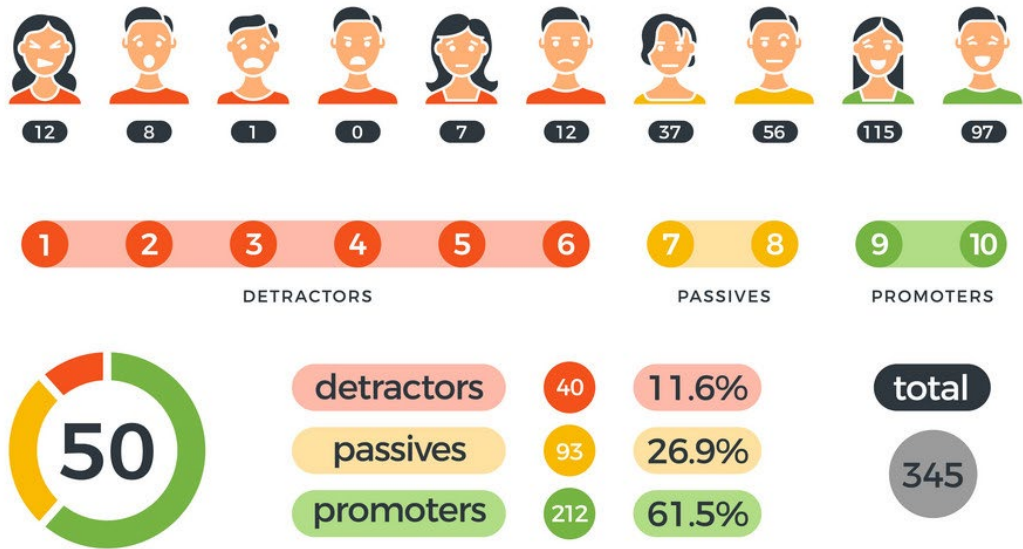


Reporting quality

Defect Trend



NET PROMOTER SCORE

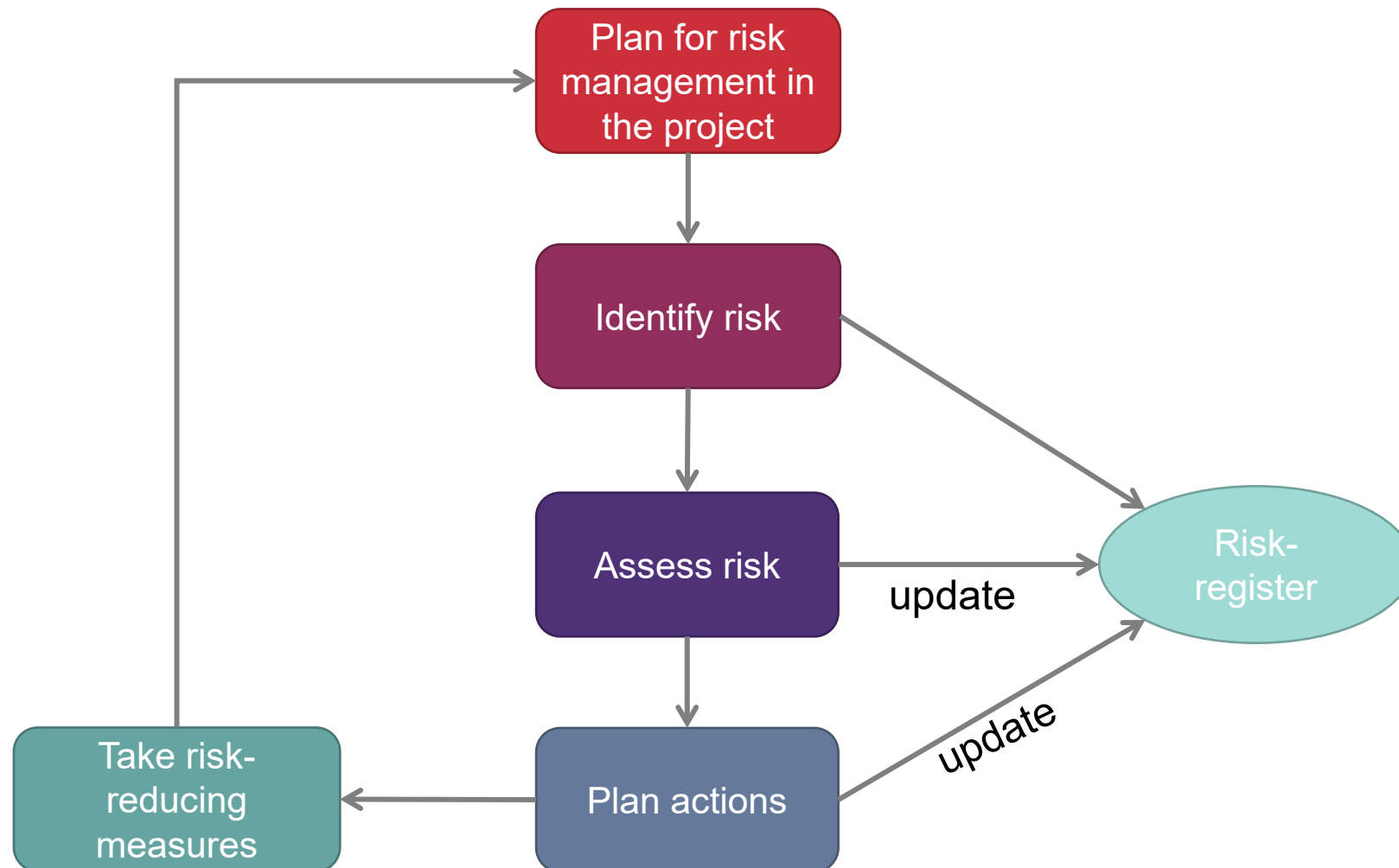


Risk management

Chapter 15

- All projects involve some sort or risk
- Source of risk include
 - the innovative nature of IT projects
 - resource availability
 - contractual relationship
 - political factors
- It is not practical to eliminate risk, rather we **manage risk**
 - Recognize the existence of risk
 - Apply methods to deal with it

Risk management in IT projects



Quantification of risk

- Risk in IT projects is often defined based on the probability of a negative event occurring combined with the severity of the consequence

Risk factor	Probability of occurring (1-5)	Severity rating (1-5)	Evaluation
Project goal and need is not well-defined	3	5	15
Changes in functional requirements	5	4	20
Project deliverables are incompletely defined	1	5	5
Project schedule is not clear	1	3	3
...			

Dealing with Risks

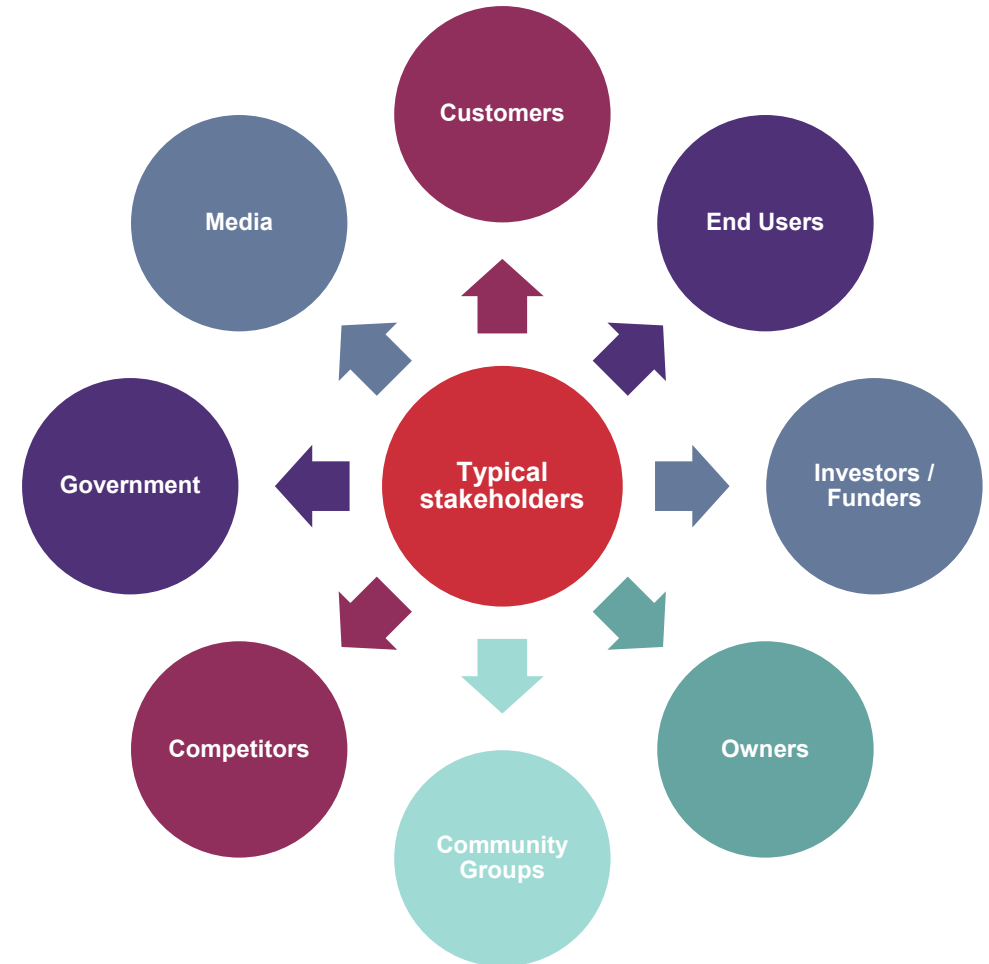
- **Avoid.** changing the project management plan to eliminate the threat entirely
- **Transfer.** shifting the negative impact of a threat, along with ownership of the response, to a third party
- **Mitigate.** reduce the probability and/or impact of an adverse risk event to be within acceptable threshold limits
- **Accept.** the project team has decided not to change the project plan to deal with a risk OR is unable to identify any other suitable response strategy

Stakeholder management and Leadership

Chapters 18 and 21

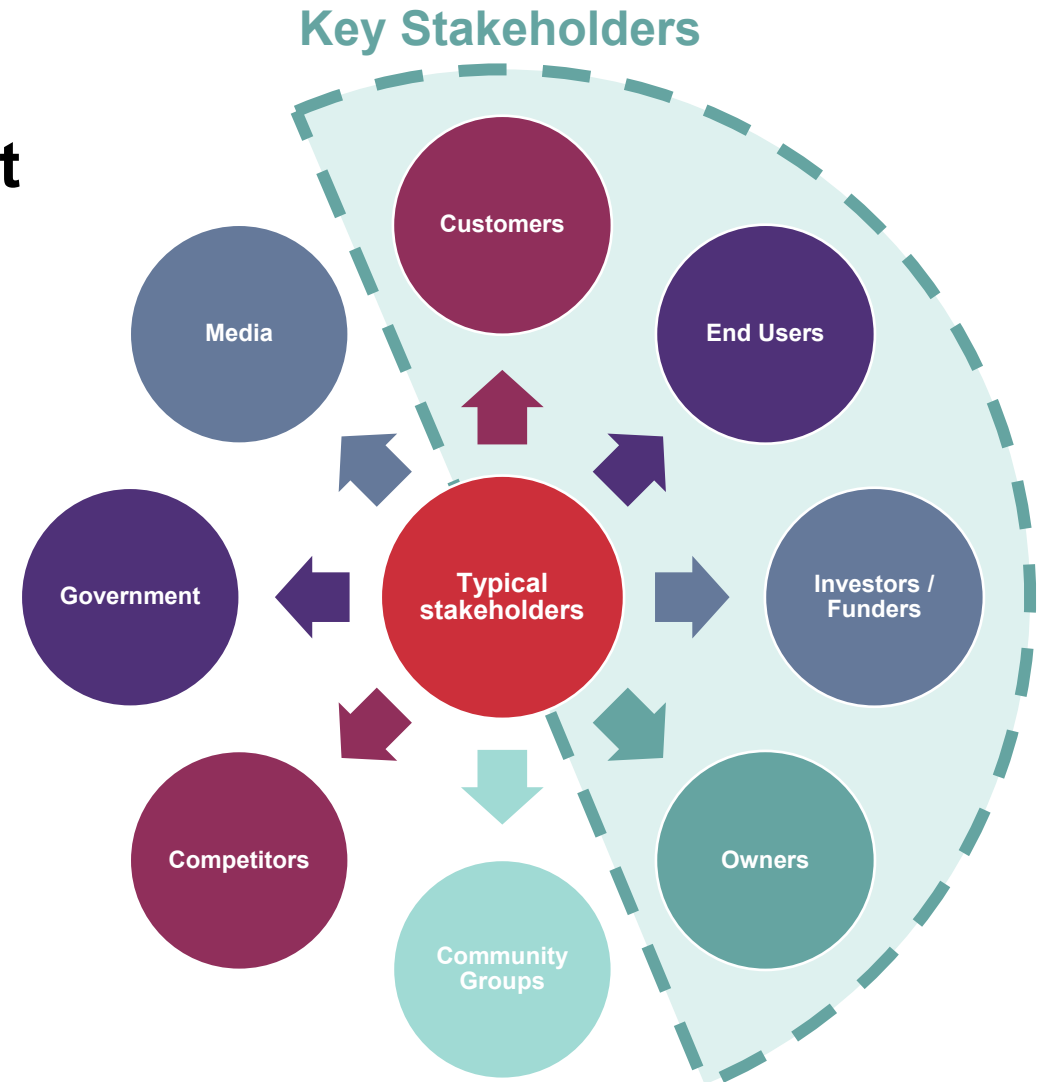
Stakeholders

- People/organizations interested in the project
- PMs should identify the stakeholders
- Plan how to involve and communicate with stakeholders



Key stakeholders

- **can influence the project to a great extent**
- have expectations for the result
- may have the power to end the project
- may have important information about user needs and the context in which a new IT system will be used



Examples of stakeholders

Developing a mobile application

Stakeholder	Interest	Power	Expectations	Engagement/communications
Project Sponsor	High	High	Clear project goals and milestones, regular progress updates	Keep informed, involve in key decision-making
End-users (customers)	High	High	User-friendly and intuitive app design, reliable and efficient app performance	Conduct user research, involve in app design and testing
Competitors	High	Low	Monitor and analyse their app features and marketing strategies	Stay informed and adapt as necessary
<i>Dev Team</i>	<i>High</i>	<i>High</i>	<i>Clear project requirements and timelines, opportunities for learning and growth</i>	<i>Involve in project planning, provide regular updates and feedback</i>
<i>Other employees</i>	<i>Low</i>	<i>Low</i>	<i>Clear project impact on their work, opportunities for involvement and feedback</i>	<i>Keep informed of project progress and how it may affect their work</i>

Leaders vs Managers

What's the difference?

Managers

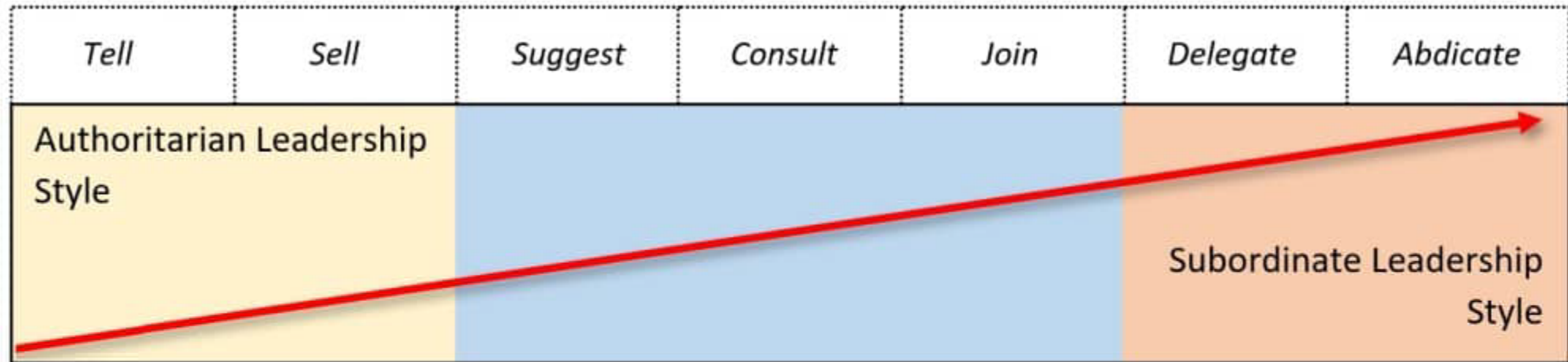
- are rather impersonal about goals: for them, goals are organizational and not personal
- coordinate and balance activities to meet conflicting schedules
- maintain a greater distance

Leaders

- take on a personal commitment and are very active about their goals.
- create excitement and change and enjoy the uncertainty.
- are emotionally involved with their people

Leadership styles

Tannenbaum and Schmidt Continuum of Leadership



1. Leader makes all decisions and announces them after the fact.

2. Leader makes all decisions and tries to sell the benefits.

3. Leader develops ideas and invites questions from others.

4. Leader presents their choice but is open to change.

5. Leader presents problems, invites suggestions, makes decision.

6. Leader sets framework then asks group to make a final decision.

7. Leader allows group choice, with only constraints being those beyond the manager's control.

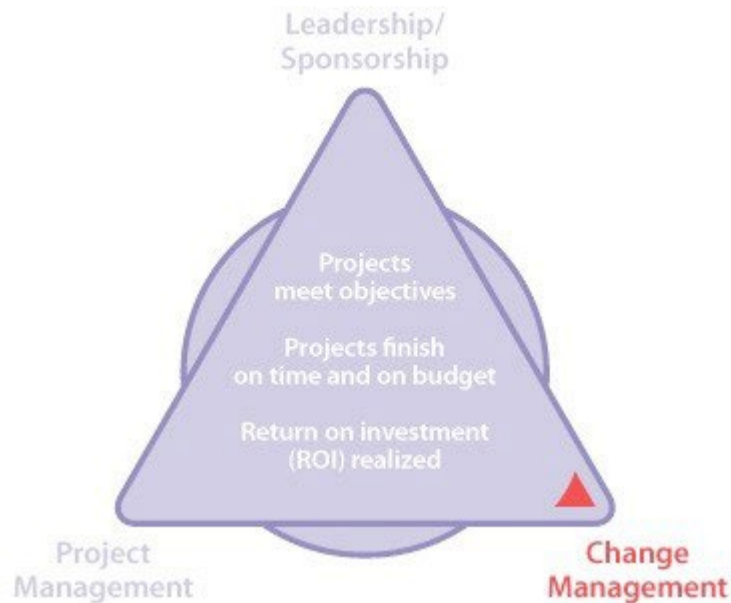
Solving conflicts

- Arrange a meeting between the parties
- Listen to what everyone has to say
- Remember that it is easy to hurt people who have worked hard for a long time
- Give people a chance – but give clear feedback and decisions
- As a project manager, you should to some extent protect team members from the noise outside the project

Change management

Chapter 20

Change management



- The process of helping transition from the **current state** to the **future state** (as defined by the change)
 - i.e. manage the **people side** of a change
- Change management requires two perspectives
 - **individual** perspective (how people experience change)
 - **organizational** perspective (how groups can be managed through a change)
- Tools like communication, sponsorship and coaching are used to help people make their own individual transition
- Change management is the final element of realizing change effectively

Change Management vs Project Management

Focus: **Technical side** of moving
from current state to future state

Project Management



Change Management

Focus: **People side** of moving
from current state to future state

Process:

- Initiation
- Planning
- Design
- Implementation
- Closure

Tools:

- Statement of work
- Project charter
- Business case
- Work breakdown structure
- Budget estimation
- Resource allocation
- Schedule
- Tracking

Process:

Organizational:

- Preparing for change
- Managing change
- Reinforcing change™

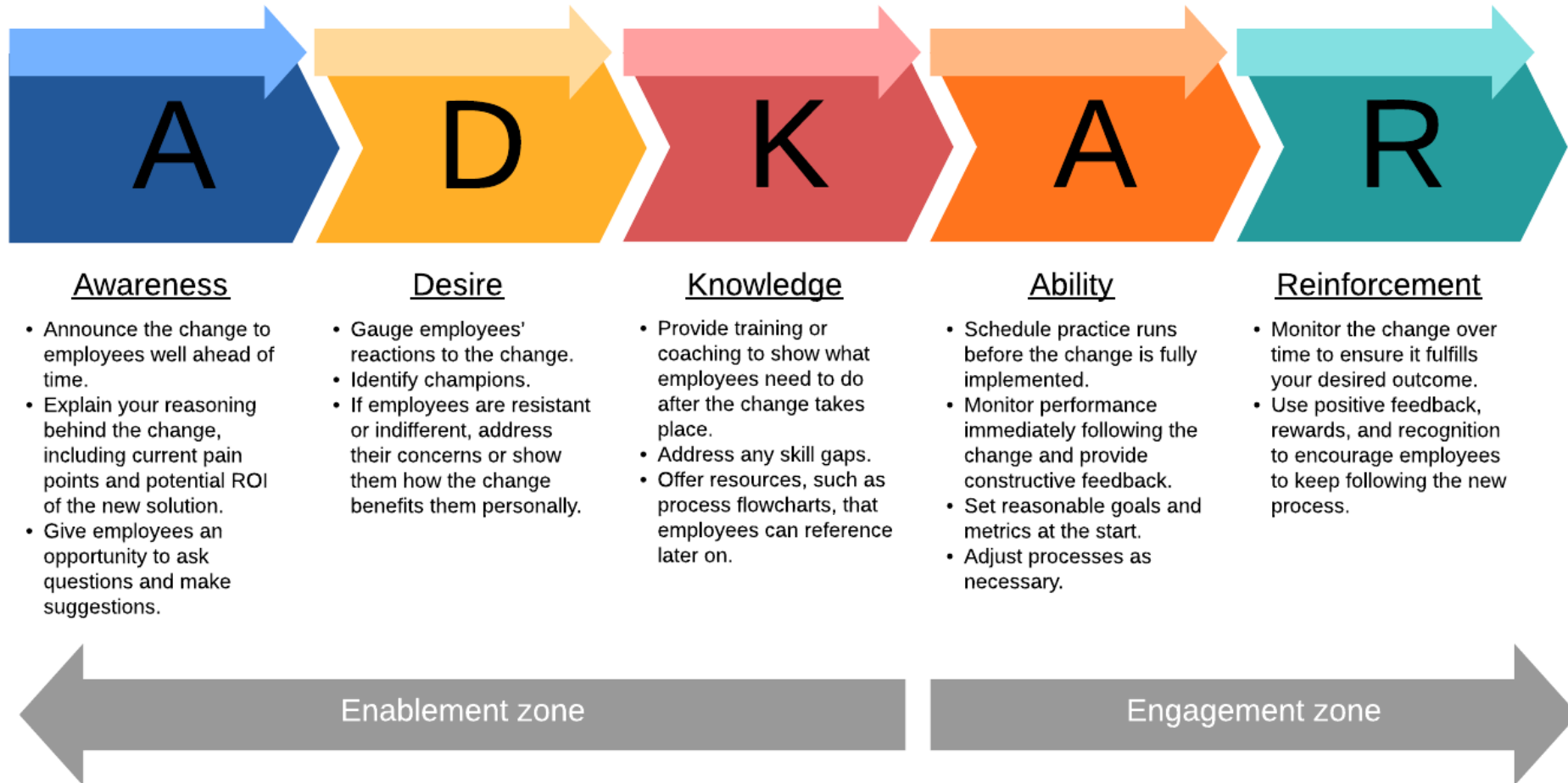
Individual:

- ADKAR®

Tools:

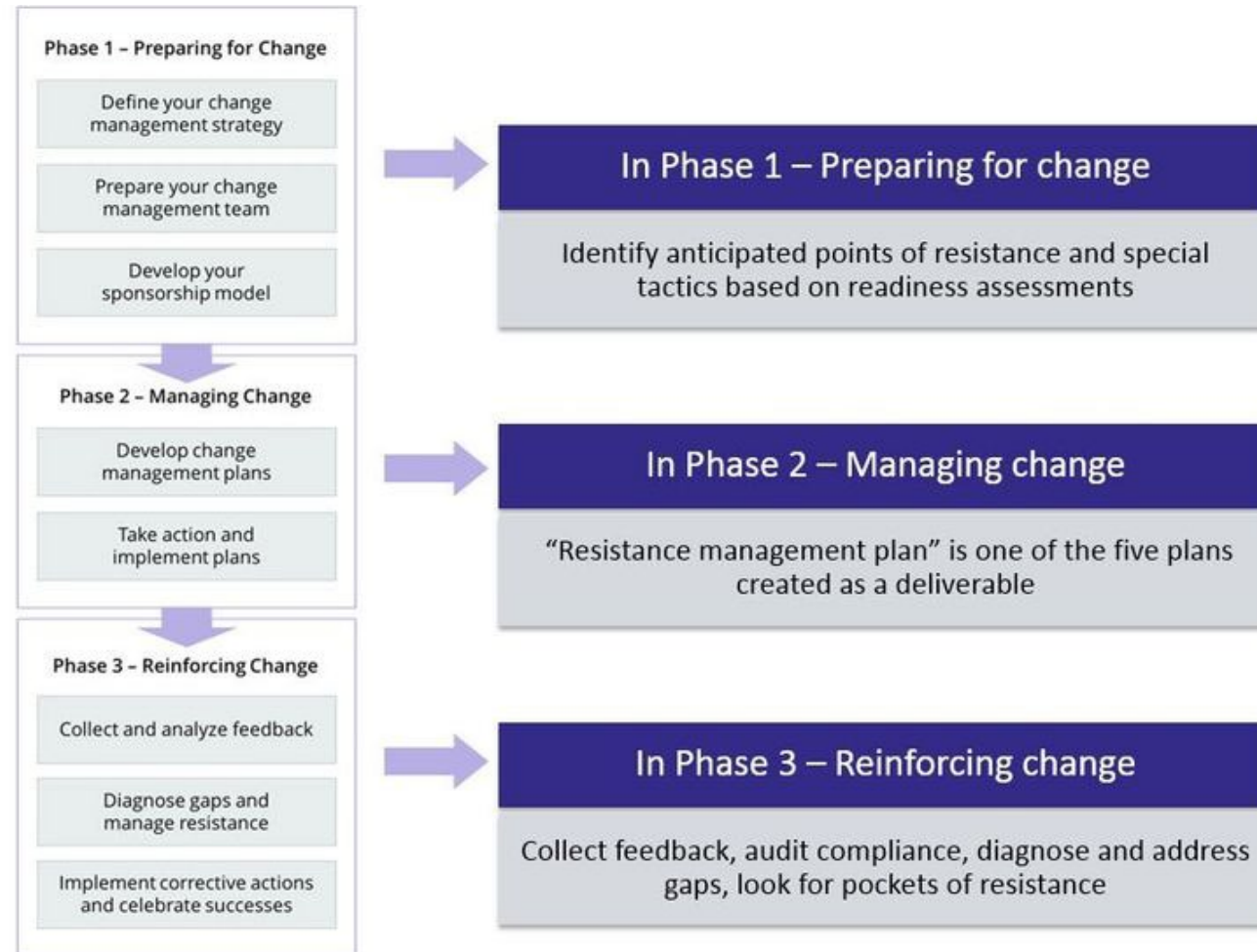
- Individual change model
- Readiness assessment
- Communications plan
- Sponsor Roadmap
- Coaching plan
- Resistance management
- Training plan
- Reinforcement

Individual Change Management



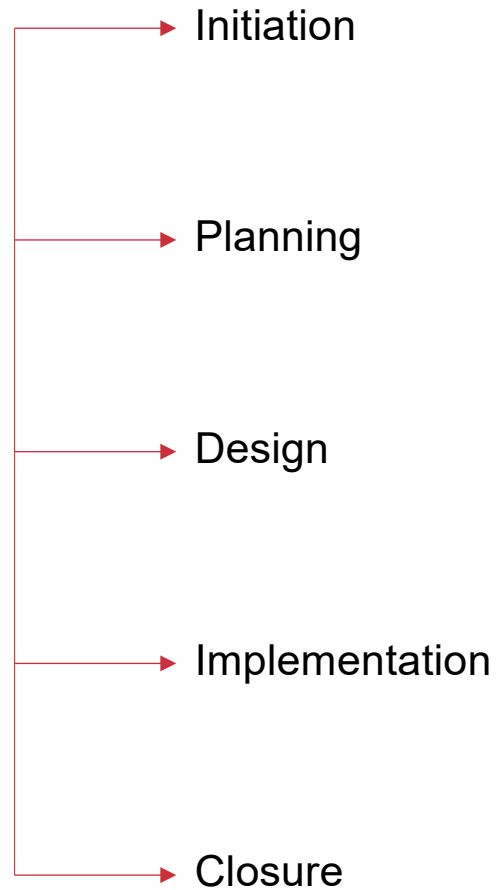
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Organizational Change Management



Entry Point Scenarios

Project Management Process



Questions?

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